Chapter Twenty

FRESNEL DIFFRACTION

One of your commissioners, M. Poisson, had deduced from the integrals reported by the author [Fresnel] the singular result that the centre of the shadow of an opaque circular screen must, when the rays penetrate there at incidences which are only a little oblique, be just as illuminated as if the screen did not exist. The consequence have been submitted to the test of a direct experiment, and observation has perfectly confirmed the calculation.

—Dominique Arago to the French Academy of Sciences*

Learning Objectives

After reading this chapter, the reader should be able to:

LO 1: analyze Fresnel half-period zones and its applications.
LO 2: describe a more rigorous approach to Fresnel diffraction.
LO 3: describe propagation of Gaussian beams.
LO 4: study diffraction by a straight edge.
LO 5: evaluate the diffraction of a plane wave by a long narrow slit and transition to the Fraunhofer region.

Important Milestones

1816 *Augustin Fresnel developed the theory of diffraction using the wave theory of light.*
1817 *Using Fresnel's theory, Poisson predicted a bright spot at the center of the shadow of an opaque disc—this is usually referred to as the 'Poisson spot'.*
1818 *Fresnel and Arago carried out the experiment to demonstrate the existence of the Poisson spot validating the wave theory.*
1874 *Marie Cornu developed a graphical approach to study Fresnel diffraction—this came to be known as the Cornu's spiral.*

20.1 INTRODUCTION

In Chapter 18, we had mentioned that the phenomenon of diffraction can be broadly classified under two categories: under the first category comes the Fresnel class of diffraction in which either the source or the screen (or both) are at a finite distance from the diffracting aperture. In the second category comes the Fraunhofer class of diffraction (discussed in the last two chapters) in which the wave incident on the aperture is a plane wave and the diffraction pattern is observed on the focal plane of a convex lens, so that the screen is effectively at an infinite distance from the aperture.

In the present chapter, we will discuss the Fresnel class of diffraction and also study the transition to the Fraunhofer region. The underlying principle in the entire analysis is the Huygens–Fresnel principle according to which:

> *Each point on a wavefront is a source of secondary disturbance and the secondary wavelets emanating from different points mutually interfere.*

In order to appreciate the implications of this principle we consider the incidence of a plane wave on a circular hole of radius a as shown in Fig. 20.1. In Sec. 18.3, we had shown that the beam will undergo diffraction divergence and the angular spreading will be given by

* The author found this quotation in Ref. 20.1.

$$\Delta\theta \sim \frac{\lambda}{2a}$$

Thus, when $a >>> \lambda$ the intensity at a point R (which is deep inside the geometrical shadow) will be negligible; on the other hand, if $a \sim \lambda$ there will be almost uniform spreading out of the beam resulting in an (almost) uniform illumina-tion of the screen. This phenomenon is a manifestation of the fact that when $a >>> \lambda$, the secondary wavelets emanating from different points on the circular aperture so beautifully interfere to produce (almost) zero intensity in the geometrical shadow and a large intensity inside the circular region (see Fig. 20.1). However, if $a \sim \lambda$ then the aperture almost acts as a point source resulting in a uniform illumination of the screen (see Fig. 12.4 and Fig. 22 in prelims pages).

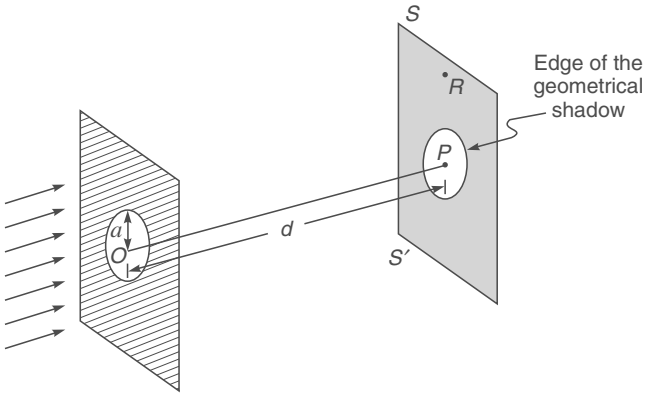


Fig. 20.1 Diffraction of a plane wave incident normally on a circular aperture of radius a.

We will first introduce the concept of Fresnel half-period zones to have a qualitative understanding of the Fresnel diffraction pattern; this will be followed by a more rigorous analysis of the Fresnel class of diffraction and its transition to the Fraunhofer region.

20.2 FRESNEL HALF-PERIOD ZONES

LO 1

Let us consider a plane wavefront WW′ propagating in the z-direction as shown in Fig. 20.2. In order to determine the field at an arbitrary point P due to the disturbances reaching from different portions of the wavefront, we make the following construction: from the point P we drop a perpendicular PO on the wavefront. If $PO = d$, then with point P as centre we draw spheres of radii $d + \lambda/2$, $d + 2\lambda/2$, $d + 3\lambda/2$,…; these spheres will intersect WW′ in circles as shown in Fig. 20.2. The radius of the nth circle will obviously be given by

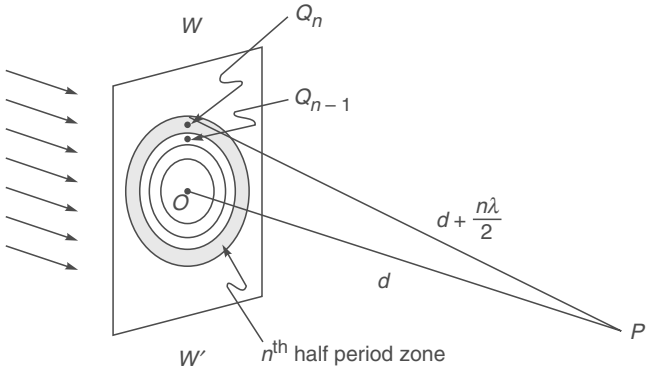


Fig. 20.2 Construction of Fresnel half-period zones.

$$r_n = \left[\left(d + n\frac{\lambda}{2}\right)^2 - d^2\right]^{1/2}$$

$$= \sqrt{n\lambda d}\left[1 + \frac{n\lambda}{4d}\right]^{1/2}$$

or

$$r_n \approx \sqrt{n\lambda d} \qquad \textbf{(20.1)}$$

where we have assumed $d >>> \lambda$, this is indeed justified for practical systems using visible light; of course, we are assuming that n is not a very large number. The annular region between the nth circle and $(n-1)^{th}$ circle is known as the nth **half-period zone**; the area of the nth half-period zone is given by

$$A_n = \pi r_n^2 - \pi r_{n-1}^2$$

$$\approx \pi[n\lambda d - (n-1)\lambda d] = \pi\lambda d \qquad \textbf{(20.2)}$$

Thus the areas of all the half-period zones are approximately equal. Now the resultant disturbance produced by the nth zone will be π out-of-phase with the disturbance produced by the $(n-1)^{th}$ [or the $(n+1)^{th}$] zone. This can easily be seen from the following consideration: For infinitesimal area surrounding a point Q_n in the nth half period zone, there is a corresponding infinitesimal area surrounding the point Q_{n-1} in the $(n-1)^{th}$ half-period zone such that

$$Q_nP - Q_{n-1}P = \frac{\lambda}{2}$$

which corresponds to a phase difference of π. Since the areas of the zones are approximately equal, one can have a one-to-one correspondence between points in various zones. Thus, the resultant amplitude at the point P can be written as

$$u(P) = u_1 - u_2 + u_3 - u_4 + \ldots + (-1)^{m+1}u_m + \ldots. \qquad \textbf{(20.3)}$$

where u_n represents the net amplitude produced by the secondary wavelets emanating from the nth zone; the alternate negative and positive signs represent the fact that the result-

ant disturbances produced by two consecutive zones are π out-of-phase with respect to each other. The amplitude produced by a particular zone is proportional to the area of the zone and inversely proportional to the distance of the zone from the point P; further, it also depends on an obliquity factor which is proportional to $\frac{1}{2}(1+\cos\chi)$ where χ is the angle that the normal to the zone makes with the line QP; this obliquity factor comes out automatically from rigorous diffraction theory.* Thus, we may write

$$u_n = \text{constant}\ \frac{A_n}{Q_n P}\frac{(1+\cos\chi)}{2} \tag{20.4}$$

where A_n represents the area of the nth zone. It can be shown that if we use the exact expression for r_n, the area of the zones increase with n; however, this slight increase in the area is exactly compensated by the increased distance of the zone from the point P. In spite of this, the amplitudes $u_1, u_2, u_3 \ldots$ decrease monotonically because of increased obliquity. Thus, we may write

$$u_1 > u_2 > u_3 \ldots \tag{20.5}$$

The series expressed by Eq. (20.3) can be approximately summed due to a method by Schuster. We rewrite Eq. (20.3) as

$$u(P) = \frac{u_1}{2} + \left(\frac{u_1}{2} - u_2 + \frac{u_3}{2}\right) + \left(\frac{u_3}{2} - u_4 + \frac{u_5}{2}\right) + \ldots \tag{20.6}$$

where the last term would either $\frac{1}{2}u_m$ or $\left(\frac{1}{2}u_{m-1} - u_m\right)$ according to m being odd or even. If the obliquity factor is such that

$$u_n > \frac{1}{2}(u_{n-1} + u_{n+1}) \tag{20.7}$$

then the quantities inside the brackets in Eq. (20.6) will be negative; consequently,

$$\left.\begin{aligned} &u(P) < \frac{1}{2}u_1 + \frac{1}{2}u_m \ ;(m \text{ odd}) \\ \text{and}\quad &u(P) < \frac{1}{2}u_1 + \frac{1}{2}u_{m-1} - u_m \approx \frac{u_1}{2} - \frac{u_m}{2} \ ;(m \text{ even}) \end{aligned}\right\} \tag{20.8}$$

where we have assumed that the amplitude of the fields produced by consecutive zones differ only slightly. In order to obtain the upper limits, we rewrite Eq. (20.3) in the form

$$u(P) = u_1 - \frac{u_2}{2} - \left(\frac{u_2}{2} - u_3 + \frac{u_4}{2}\right) - \left(\frac{u_4}{2} - u_5 + \frac{u_6}{2}\right) \tag{20.9}$$

where the last term would now be $\left(-\frac{1}{2}u_{m-1} + u_m\right)$ when m is odd and $-\frac{1}{2}u_m$ when m is even. Since the quantities inside the brackets are negative, we obtain

$$\left.\begin{aligned} &u(P) > u_1 - \frac{u_2}{2} - \frac{u_{m-1}}{2} + u_m \approx \frac{u_1}{2} + \frac{u_m}{2} \\ &\qquad\qquad (\text{when } m \text{ is odd}) \\ \text{and}\quad &u(P) > u_1 - \frac{u_2}{2} - \frac{u_m}{2} \approx \frac{u_1}{2} - \frac{u_m}{2} \\ &\qquad\qquad (\text{when } m \text{ is even}) \end{aligned}\right\} \tag{20.10}$$

Using Eqs. (20.8) and (20.10) we may approximately write

$$\left.\begin{aligned} &u(P) \approx \frac{u_1}{2} + \frac{u_m}{2}\ (\text{when } m \text{ is odd}) \\ \text{and}\quad &u(P) \approx \frac{u_1}{2} - \frac{u_m}{2}\ (\text{when } m \text{ is even}) \end{aligned}\right\} \tag{20.11}$$

If we can neglect u_m in comparison to u_1 then the Eq. (20.11)** gives the remarkable result that

$$u(P) \approx \frac{u_1}{2} \tag{20.12}$$

implying that *the resultant amplitude produced by the entire wavefront is only one half of the amplitude produced by the first half-period zone.*

20.2.1 Diffraction by a Circular Aperture

We may use the above analysis to study the diffraction of a plane wave by a circular aperture. Let the point P be at a distance d from the circular aperture (see Fig. 20.1). We assume that the radius of the circular aperture, a, can be increased from zero onwards. As a increases, the intensity at the point P would also increase till the circular aperture contains the first half-period zone; this would happen when $a = \sqrt{\lambda d}$. The resultant amplitude at the point P would be u_1 which is twice the value of the amplitude for the unobstructed wavefront [see Eq. (20.12)]. The intensity would therefore be $4I_0$, where I_0 represents the intensity at the point P due to the unobstructed wavefront. If we further increase

* See, e.g., Ref. 20.3.

** If one assumes a form of the obliquity factor as given by Eq. (20.4) then it decreases from 1 to 1/2 as m increases from 1 to ∞; this implies that $|u_m|$ can never be smaller that $\frac{u_1}{2}$. However, when m is large, a slight shift of the point P on the axis will change the amplitude from $\frac{u_1}{2} + \frac{u_m}{2}$ to $\frac{u_1}{2} - \frac{u_m}{2}$, the changes will occur with such great rapidity that one can only observe the average value which will be $\frac{u_1}{2}$.

a then $u(P)$ would start decreasing and when the circular aperture contains the first two half-period zones (which would happen when $a = \sqrt{2\lambda d}$) the resultant amplitude ($= u_1 - u_2$) would be almost zero. *Thus, by increasing the hole diameter, the intensity at the point P decreases almost to zero*. This interesting result is once again due to the validity of the Huygens–Fresnel principle and hence would be valid for sound waves also. We may generalize the above result by noting that if

$$a = \sqrt{(2n+1)\lambda d}\ ; n = 0, 1, 2,\ldots \text{ [maxima]}$$

the aperture will contain an odd number of half-period zones and the intensity will be maximum; on the other hand, if

$$a = \sqrt{2n\lambda d}\ ; n = 1, 2,\ldots \text{ [minima]}$$

the aperture will contain an even number of half-period zones and the intensity will be minimum. In order to have a numerical appreciation, we note that for $d = 50$ cm and $\lambda = 5 \times 10^{-5}$ cm, the radii of the first, second and third zones would be 0.500 mm, 0.707 mm and 0.866 mm, respectively. As a corollary of the above analysis we can consider a circular aperture of a fixed radius a and study the intensity variation along the axis. Whenever the distance

$$d = \frac{a^2}{(2n+1)\lambda}\ ; n = 0, 1, 2,\ldots \text{ (maxima)}$$

the point P (see Fig. 20.1) will correspond to a maximum. Similarly, when

$$d = \frac{a^2}{2n\lambda}\ ; n = 1, 2,\ldots \text{ (minima)}$$

the point P will correspond to a minimum. The intensity distribution on a screen SS' at off-axis points can be approximately calculated by using the half-period zones, but such a calculation is fairly cumbersome. However, from the symmetry of the problem, one can deduce that the diffraction pattern has to be in the form of concentric circular rings with their centres at the point P.

20.2.2 Diffraction by an Opaque Disc–The Poisson Spot

If instead of the circular aperture we have a circular disc [see Fig. 20.3(a)] and if the disc obstructs the first p half-period zones then the field at the point P would be

$$u(P) = u_{p+1} - u_{p+2} + \ldots \approx \frac{u_{p+1}}{2} \qquad \textbf{(20.13)}$$

Thus, we should always obtain a bright spot on the axis behind a circular disc; (the more rigorous theory also predicts the same result — see Sec. 20.4.2). This is called the '**Poisson spot**'. We may mention here that it was in 1816 that the French physicist Augustin Fresnel developed the mathematical theory of diffraction using the wave theory of light. Simeon Poisson, the famous mathematician, used Fresnel's theory to predict a bright spot at the center of the shadow of an opaque disc. Poisson was a great supporter of the corpuscular theory of light and he said that since the bright spot is against common sense, the wave theory must be wrong. Shortly afterwards, Fresnel and Arago carried out the experiment to demonstrate the existence of the Poisson spot [see Fig. 20.3(b)], validating the wave theory.

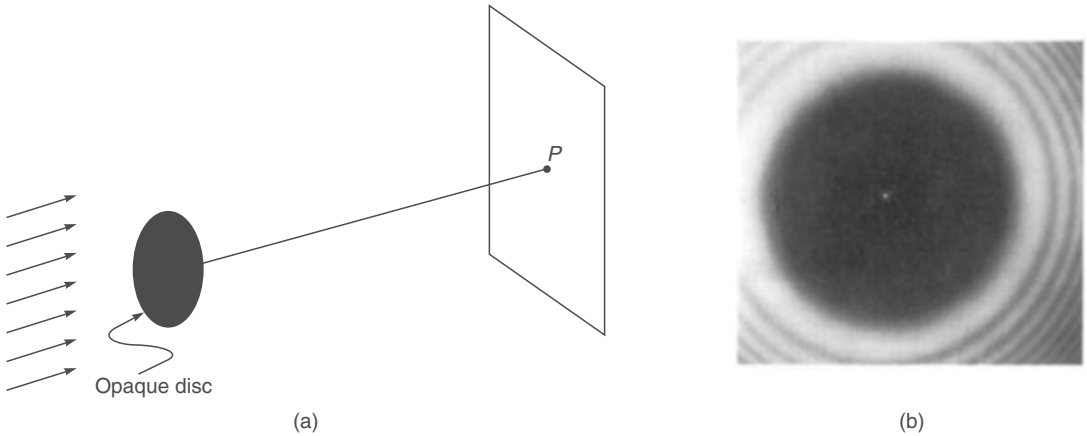


Fig. 20.3 (a) When a plane wave is incident normally on an opaque disc, a bright spot is always formed on an axial point. This spot is known as the Poisson spot. (b) The Poisson spot at the center of the shadow of a one cent coin; the screen is 20 m from the coin and the source of light is also 20 m from the coin. [Photograph reprinted with permission from P.M. Rinard, *Large scale diffraction patterns from circular objects*, American Journal of Physics, Vol 44, 70, 1976; Copyright (1976), American Association of Physics Teachers used with Permission].

20.3 THE ZONE-PLATE LO 1

A beautiful application of the concept of Fresnel half-period zones lies in the construction of the zone-plate which consists of a large number of concentric circles whose radii are proportional to the square root of natural numbers and the alternate annular regions of which are blackened (see Fig. 20.4). Let the radii of the circles be $\sqrt{1}\,K$, $\sqrt{2}\,K$, $\sqrt{3}\,K$, $\sqrt{4}\,K$,… where K is a constant and has the dimension of length. We consider a point P_1 which is at a distance K^2/λ from the zone plate; for this point the blackened rings correspond to the 2nd, 4th, 6th, … half-period zones. Thus, the even zones are obstructed and the resultant amplitude at P_1 [see Fig. 20.5 (a)] will be

$$u_1 + u_3 + u_5 + \ldots \qquad \textbf{(20.14)}$$

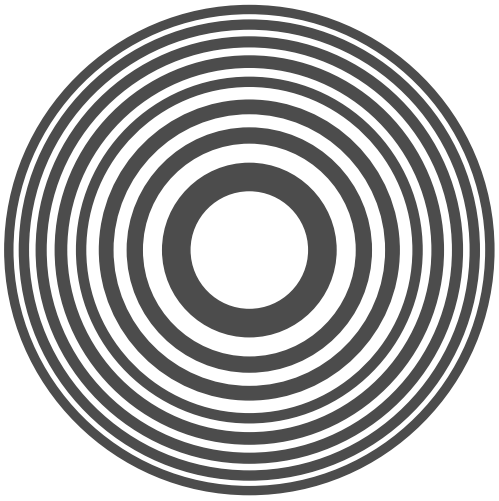


Fig. 20.4 The zone plate.

producing an intense maximum. For the point P_3 (which is at a distance $K^2/3\lambda$) the first blackened ring contains the 4th, 5th, 6th zones, the second blackened ring contains the 10th, 11th and 12th zones, etc.; thus the resultant amplitude would be

$$(u_1 - u_2 + u_3) + (u_7 - u_8 + u_9) + \ldots \qquad \textbf{(20.15)}$$

which would again correspond to a maximum, but it would not be as intense as the point P_1. Between the points P_1 and P_3 there will be a point P_2 (at a distance $K^2/2\lambda$) where the resultant amplitude would be

$$(u_1 - u_2) + (u_5 - u_6) + \ldots \qquad \textbf{(20.16)}$$

implying that corresponding to P_2 the first blackened ring contains the 3rd and 4th half-period zones, etc. Obviously, the point P_2 will correspond to a minimum. Thus, if a plane wave is incident normally on a zone-plate, then the corresponding focal points are at distances

$$\frac{K^2}{\lambda}, \frac{K^2}{3\lambda}, \frac{K^2}{5\lambda}, \ldots \qquad \textbf{(20.17)}$$

from the zone-plate. Elementary calculations will show that the zone-plate suffers from considerable chromatic aberrations [see Problem 20.5].

Example 20.1 Assume a plane wave ($\lambda = 5 \times 10^{-5}$ cm) to be incident on a circular aperture of radius 0.5 mm. We will calculate the positions of the brightest and darkest points on the axis. For the brightest point, the aperture should contain only the first zone and thus we must have (see Fig. 20.1)

$$(0.05)^2 = OP\,(5 \times 10^{-5})$$

Thus OP = 50 cm. Similarly, the darkest point would be at a distance

$$\frac{(0.05)^2}{2 \times 5 \times 10^{-5}} = 25 \text{ cm}$$

Example 20.2 Consider a zone-plate with radii

$$r_n = 0.1\sqrt{n} \text{ cm}$$

For $\lambda = 5 \times 10^{-5}$ cm, we will calculate the positions of various foci. The most intense focal point will be at a distance

$$\frac{r_1^2}{\lambda} = \frac{0.01}{5 \times 10^{-5}} = 200 \text{ cm}$$

The other focal points will be at distances of 200/3, 200/5, 200/7 cm, etc. Between any two consecutive foci there will be dark points on the axis corresponding to which the first circle will contain an even number of half-period zones.

The zone-plate can also be used for imaging points on the axis, e.g., if we have a point source at S then a bright image will be formed at P, where the point P should be such that (see Fig. 20.5(b)):

$$SL + LP - SP = \frac{\lambda}{2} \qquad \textbf{(20.18)}$$

the point L being on the periphery of the first circle of the zone plate [see Fig. 20.5(b)]. If the radius of the first circle is r_1, then

$$\begin{aligned} SL + LP - SP &= \sqrt{a^2 + r_1^2} + \sqrt{b^2 + r_1^2} - (a + b) \\ &\approx a\left[1 + \frac{r_1^2}{2a^2}\right] + b\left[1 + \frac{r_1^2}{2b^2}\right] - (a + b) \\ &\approx \frac{r_1^2}{2}\left(\frac{1}{a} + \frac{1}{b}\right) \end{aligned} \qquad \textbf{(20.19)}$$

Thus Eq. (20.18) becomes

$$\frac{1}{a} + \frac{1}{b} = \frac{1}{f} \qquad \textbf{(20.20)}$$

where $f = r_1^2/\lambda$ represents the focal length. Equation (20.20) resembles the lens law. A very interesting demonstration experiment of the zone-plate can be carried out by using

microwave sources ($\lambda \sim 1$ cm) and, instead of the dark rings, having aluminum rings on a perspex sheet of dimension ~ 40 cm $\times$ 40 cm.

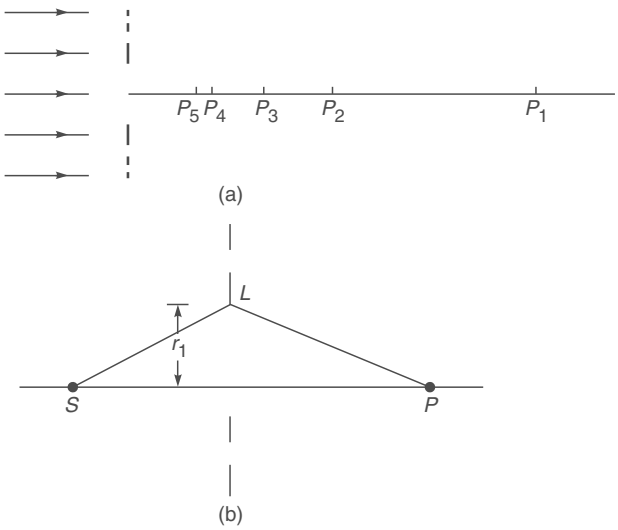


Fig. 20.5 (a) For a plane wave incident on a zone-plate, the maximum intensity occurs at the points P_1, P_3 etc.; the minima occur at P_2, P_4, ... (b) Imaging of a point object by a zone-plate.

20.4 FRESNEL DIFFRACTION—A MORE RIGOROUS APPROACH

LO 2

In Sec. 19.2, we had considered a plane wave (of amplitude A) incident normally on an aperture as shown in Fig. 20.6. Using Huygens–Fresnel principle we had shown that the field produced at the point P on a screen SS' (which is at a distance d from the aperture) is given by

$$u(P) = \frac{A}{i\lambda}\iint \frac{e^{ikr}}{r} d\xi\, d\eta \qquad \textbf{(20.21)}$$

where the integration is over the area of the aperture. Now, if the amplitude and phase distribution on the plane $z = 0$ is given by $A(\xi, \eta)$ then the above integral is modified to

$$u(P) = \frac{1}{i\lambda}\iint A(\xi, \eta)\frac{e^{ikr}}{r} d\xi\, d\eta \qquad \textbf{(20.22)}$$

Further, in the Fresnel approximation [see Eq. 19.9] the above integral takes the form

$$u(x, y, z) \approx \frac{1}{i\lambda z} e^{ikz} \iint A(\xi, \eta) \times \exp\left\{\frac{ik}{2z}[x - \xi)^2 + (y - \eta)^2]\right\} d\xi\, d\eta \qquad \textbf{(20.23)}$$

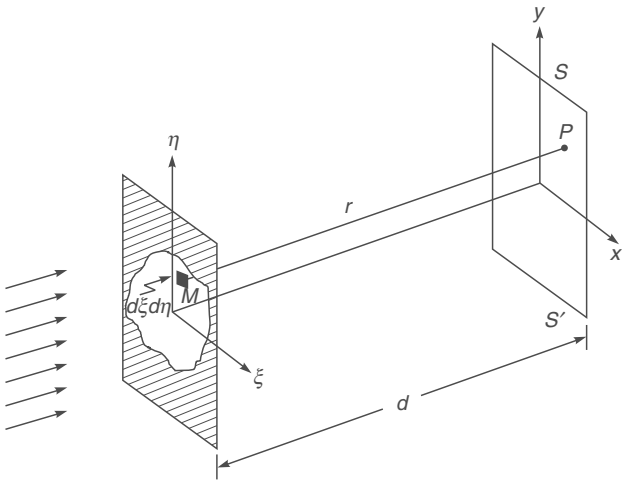


Fig. 20.6 A plane wave incident normally on an aperture.

20.4.1 Diffraction of a Plane Wave Incident Normally on a Circular Aperture

We assume a plane wave incident normally on a circular aperture of radius a as shown in Fig. 20.7. The z-axis is normal to the plane of the aperture and the screen SS' is assumed to be normal to the z-axis. It is obvious from the symmetry of the problem that we will obtain circular fringes on the screen SS'; however, it is very difficult to calculate the actual intensity variation on the screen. Therefore, for the sake of

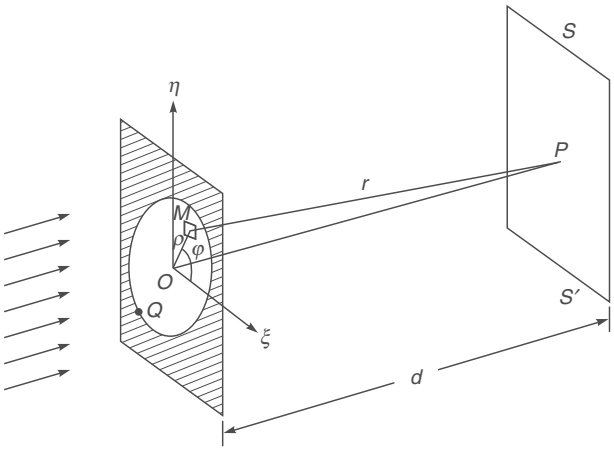


Fig. 20.7 Diffraction of a plane wave incident normally on a circular aperture of radius a; Q is an arbitrary point on the periphery of the aperture.

mathematical simplicity, we will calculate the variation of intensity only along the z-axis. Obviously, it will be more convenient to use the circular system of coordinates. In this system, the coordinates of an arbitrary point M on the aperture will be (ρ, ϕ) where ρ is the distance of the point M from the center O and ϕ is the angle that OM makes with the ξ-axis [see Fig. 20.7] and a small element area dS surrounding the point M will be $\rho\, d\rho\, d\phi$. Thus,

$$u(P) \approx -\frac{i\,A}{\lambda}\int_0^{2\pi}\int_0^{a}\frac{e^{ikr}}{r}\,\rho\, d\rho\, d\phi \qquad \textbf{(20.24)}$$

Now,

$$\rho^2 + d^2 = r^2$$

Thus,

$$\rho\, d\rho = r\, dr$$

and Eq. (20.24) becomes

$$u(P) \approx -\frac{i\,A}{\lambda}\int_0^{2\pi}\int_d^{\sqrt{a^2+d^2}} e^{ikr}\, dr\, d\phi \qquad \textbf{(20.25)}$$

The integration is very simple and since $k = 2\,\pi/\lambda$, we readily obtain

$$u(P) \approx A\, e^{ikd}\,(1 - e^{ip\pi}) \qquad \textbf{(20.26)}$$

where we have defined p by the following equation

$$k\left[\sqrt{a^2 + d^2} - d\right] = p\pi$$

The above equation implies

$$QP - OP = \frac{p\,\lambda}{2}$$

where Q is a point on the periphery of the circular aperture (see Fig. 20.7). From Eq. (20.26) we readily get

$$I(P) = 4\, I_0 \sin^2\frac{p\,\pi}{2} \qquad \textbf{(20.27)}$$

where I_0 is the intensity associated with the incident plane wave. Equation (20.27) tells us that the intensity is zero or maximum when p is an even or odd integer, i.e., when $QP - OP$ is an even or odd multiple of $\lambda/2$. This can be understood physically by using the concept of Fresnel half-period zones discussed in Sec. 20.2. Thus, if the aperture contains an even number of half-period zones, the intensity at the point P will be negligibly small and conversely, if the circular aperture contains an odd number of zones, the intensity at the point P will be maximum. Now, when $d << a$ (as is usually the case)

$$p \approx \frac{k}{\pi}\left[d\left(1 + \frac{a^2}{2d^2}\right) - d\right]$$

or

$$p \approx \frac{a^2}{\lambda d} \qquad \textbf{(20.28)}$$

which is known as the **Fresnel number** of the aperture. In Fig. 20.8, we have plotted the corresponding intensity variation as a function of the dimensionless parameter

$$\frac{\lambda d}{a^2}$$

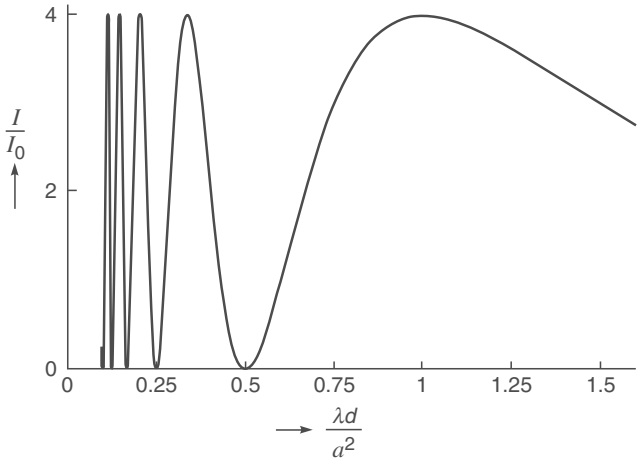


Fig. 20.8 The intensity variation on an axial point corresponding to a plane wave incident on a circular aperture of radius a.

The figure shows that when the (circular) aperture contains an even number of half-period zones, the intensity at the point P will be zero and when the aperture contains an odd number of zones, the intensity at the point P will be maximum.

20.4.2 Diffraction by a Circular Disc

We next consider the diffraction pattern produced by an opaque disc of radius a (see Fig. 20.3). Once again we will assume that the observation point lie on the axis of the disc. Equation (20.21) tells us that in order to calculate the field we have to carry out an integration over the open region of the aperture. Obviously, if $u_1(P)$ and $u_2(P)$ respectively represent the fields at the point P due to a circular aperture and an opaque disc (of the same radius), then

$$u_1(P) + u_2(P) = u_0(P) \qquad \textbf{(20.29)}$$

where $u_0(P)$ represents the field in the absence of any aperture; Eq. (20.29) is known as the **Babinet's principle**. Thus,

$$u_2(P) = u_0(P) - u_1(P)$$
$$= u_0(P) - u_0(P)[1 - e^{ip\pi}]$$

or
$$u_2(P) = u_0(P)\, e^{ip\pi} \quad \textbf{(20.30)}$$

where, for $u_1(P)$ we have used Eq. (20.26). Thus the intensity at the point P on the axis of a circular disc would be

$$I_2(P) = |u_2(P)|^2 = I_0(P) \quad \textbf{(20.31)}$$

which gives us the remarkable result that the intensity at a point on the axis of an opaque disc is equal to the intensity at the point in the absence of the disc! This is the Poisson spot discussed in Sec. 20.2.2.

20.5 GAUSSIAN BEAM PROPAGATION

LO 3

When a laser oscillates in its fundamental transverse mode, the transverse amplitude distribution is Gaussian. Also, the output of a single mode fiber is very nearly Gaussian. Therefore, the study of the diffraction of a Gaussian beam is of great importance. We assume a Gaussian beam propagating along the z-direction whose amplitude distribution on the plane $z = 0$ is given by

$$A(\xi, \eta) = a \exp\left[-\frac{\xi^2 + \eta^2}{w_0^2}\right] \quad \textbf{(20.32)}$$

implying that the phase front is plane at $z = 0$. From Eq. (20.32), it follows that at a distance w_0 from the z-axis, the amplitude falls by a factor $1/e$ (i.e., the intensity reduces by a factor $1/e^2$). This quantity w_0 is called the *spot size* of the beam. Substituting Eq. (20.32) in Eq. (20.23) and carrying out the integration, we obtain [see Appendices D and E]

$$u(x, y, z) \approx \frac{a}{\left(1 + \frac{iz}{\sqrt{\alpha}}\right)} \exp\left[-\frac{x^2 + y^2}{w^2(z)}\right] e^{i\Phi} \quad \textbf{(20.33)}$$

where

$$\alpha = \frac{\pi^2 w_0^4}{\lambda^2} \quad \textbf{(20.34)}$$

$$w(z) = w_0 \sqrt{1 + \frac{z^2}{\alpha}} \quad \textbf{(20.35)}$$

$$\Phi = kz + \frac{k}{2R(z)}(x^2 + y^2) \quad \textbf{(20.36)}$$

$$R(z) = z\left[1 + \frac{\pi^2 w_0^4}{\lambda^2 z^2}\right] = z + \frac{\alpha}{z} \quad \textbf{(20.37)}$$

Thus, the intensity distribution is given by

$$I(x, y, z) = \frac{I_0}{\left(1 + \frac{z^2}{\alpha}\right)} \exp\left[-\frac{2(x^2 + y^2)}{w^2(z)}\right] \quad \textbf{(20.38)}$$

which show that the transverse intensity distribution remains Gaussian with the beam-width increasing with z which essentially implies diffraction divergence. In Sec. 18.4, we had shown that

(i) For a given value of λ, the diffraction divergence increases as we decrease the value of the initial spot size (see Fig. 18.13).

(ii) For a given value of w_0, the diffraction divergence decreases with decrease in the value of λ (see Fig. 18.14).

Now, for a spherical wave *diverging* from the origin, the field distribution is given by

$$u \sim \frac{1}{r} e^{ikr} \quad \textbf{(20.39)}$$

On the plane $z = R$ (see Fig. 20.9)

$$r = \sqrt{x^2 + y^2 + R^2} = R\sqrt{1 + \frac{x^2 + y^2}{R^2}}$$
$$\approx R + \frac{x^2 + y^2}{2R}$$

where we have assumed $|x|, |y| << R$. Thus on the plane $z = R$, the phase distribution (corresponding to a spherical wave of radius R) would be given by

$$e^{ikr} \approx e^{ikR}\, e^{\frac{ik}{2R}(x^2 + y^2)} \quad \textbf{(20.40)}$$

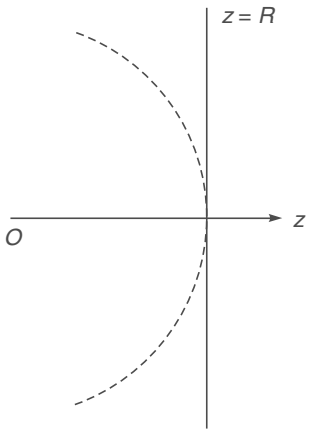


Fig. 20.9 A spherical wave diverging from the point O. The dashed curve represents a section of the spherical wavefront at a distance R from the source.

From the above equation it follows that a phase variation of the type

$$\exp\left[i\frac{k}{2R}(x^2+y^2)\right]$$

(on the x-y plane) represents a *diverging* spherical wave of radius R. If we now compare the above expression with Eqs. (20.36) and (20.37) we can say that the radius of curvature of the phase front of the propagating Gaussian beam is given by Eq. (20.37). Thus as the beam propagates, the phase front which was plane at $z = 0$ becomes curved. In Sec. 27.5, we will use the above analysis to determine conditions for a Gaussian beam to resonate between two spherical mirrors. Here, as a simple application, we will consider a resonator configuration consisting of a plane mirror and a spherical mirror separated by a distance d (see Fig. 20.10); indeed such a configuration is used to produce a single transverse mode oscillation in a ruby laser. On the plane $z = 0$, the phase front is plane and for the beam to resonate, the phase front must have a radius of curvature equal to the radius of curvature of the mirror:

$$R = d\left[1+\frac{\pi^2 w_0^4}{\lambda^2 d^2}\right] \quad \Rightarrow \quad w_0^2 = \frac{\lambda d}{\pi}\sqrt{\left(\frac{R}{d}-1\right)} \qquad \textbf{(20.41)}$$

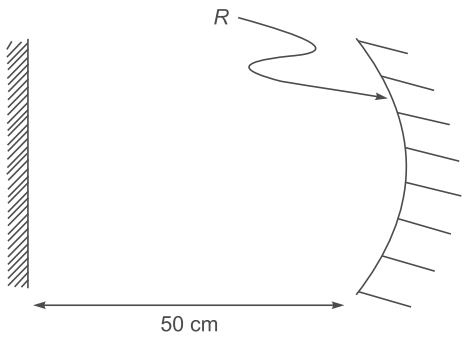


Fig. 20.10 A simple resonator consisting of a plane mirror and a concave mirror of radius R.

Example 20.3 We assume $\lambda \approx 0.6328\ \mu$m, $d \approx 50$ cm and $R \approx 100$ cm; which are typical parameters for a He-Ne laser. Simple calculations will show that $w_0 \approx 0.32$ mm. If we increase R to 200 cm, we will get $w_0 \approx 0.38$ mm. Obviously R should be greater than d. In Sec. 27.5, we will consider the more general case of a Gaussian beam resonating between two mirrors of radii of curvatures R_1 and R_2 as shown in Fig. 27.26.

20.6 DIFFRACTION AT A STRAIGHT EDGE LO 4

Before we discuss the straight edge diffraction pattern, we introduce the Fresnel integrals.

Fresnel Integrals: Fresnel integrals are defined by the following equations:

$$C(\tau) = \int_0^{\tau} \cos\left(\frac{1}{2}\pi u^2\right) du \qquad \textbf{(20.42)}$$

and

$$S(\tau) = \int_0^{\tau} \sin\left(\frac{1}{2}\pi u^2\right) du \qquad \textbf{(20.43)}$$

Since the integrands are even functions of τ, the Fresnel integrals $C(\tau)$ and $S(\tau)$ are odd functions of τ:

$$C(-\tau) = -C(\tau) \quad \text{and} \quad S(-\tau) = -S(\tau) \qquad \textbf{(20.44)}$$

Further, since

$$\int_{-\infty}^{\infty} e^{-\alpha x^2}\, dx = \sqrt{\frac{\pi}{\alpha}} \qquad \textbf{(20.45)}$$

we have

$$\int_{-\infty}^{\infty} e^{i\pi u^2/2}\, du = \sqrt{\frac{\pi}{-i\pi/2}} = \sqrt{2}\, e^{i\pi/4} = (1+i) \qquad \textbf{(20.46)}$$

Now,

$$\int_{-\infty}^{+\infty} \exp\left[i\frac{\pi u^2}{2}\right] du$$

$$= 2\left[\int_0^{\infty} \cos\left(\frac{1}{2}\pi u^2\right) du + i\int_0^{\infty} \sin\left(\frac{1}{2}\pi u^2\right) du\right]$$

$$= 2\,[C(\infty) + i\,S(\infty)]$$

Thus, using Eq. (20.46), we get $C(\infty) = \frac{1}{2} = S(\infty)$.

To summarize, the Fresnel integrals have the following important properties:

$$C(\infty) = S(\infty) = \frac{1}{2};\ C(0) = S(0) = 0 \qquad \textbf{(20.47)}$$

$$C(-\tau) = -C(\tau) \quad \text{and} \quad S(-\tau) = -S(\tau) \qquad \textbf{(20.48)}$$

The values of the Fresnel integrals for typical values of τ are tabulated in Table 20.1.

Table 20.1 Table of Fresnel Integrals*

$$C(\tau)=\int_0^{\tau}\cos\left(\frac{\pi}{2}v^2\right)dv;\quad S(\tau)=\int_0^{\tau}\sin\left(\frac{\pi}{2}v^2\right)dv$$

τ	$C(\tau)$	$S(\tau)$	τ	$C(\tau)$	$S(\tau)$
0.0	0.00000	0.00000	2.6	0.38894	0.54999
0.2	0.19992	0.00419	2.8	0.46749	0.39153
0.4	0.39748	0.03336	3.0	0.60572	0.49631
0.6	0.58110	0.11054	3.2	0.46632	0.59335
0.8	0.72284	0.24934	3.4	0.43849	0.42965
1.0	0.77989	0.43826	3.6	0.58795	0.49231
1.2	0.71544	0.62340	3.8	0.44809	0.56562
1.4	0.54310	0.71353	4.0	0.49843	0.42052
1.6	0.36546	0.63889	4.2	0.54172	0.56320
1.8	0.33363	0.45094	4.4	0.43833	0.46227
2.0	0.48825	0.34342	4.6	0.56724	0.51619
2.2	0.63629	0.45570	4.8	0.43380	0.49675
2.4	0.55496	0.61969	5.0	0.56363	0.49919
			∞	0.5	0.5

Figure 20.11 gives a parametric representation of the Fresnel integrals and is known as the **Cornu's spiral**. The horizontal and the vertical axes represent $C(\tau)$ and $S(\tau)$, respectively and the numbers written on the spiral are the values of τ. For example, as can be seen from the figure, for $\tau = 1.0$, $C(\tau) \approx 0.77989$ and $S(\tau) \approx 0.43826$.

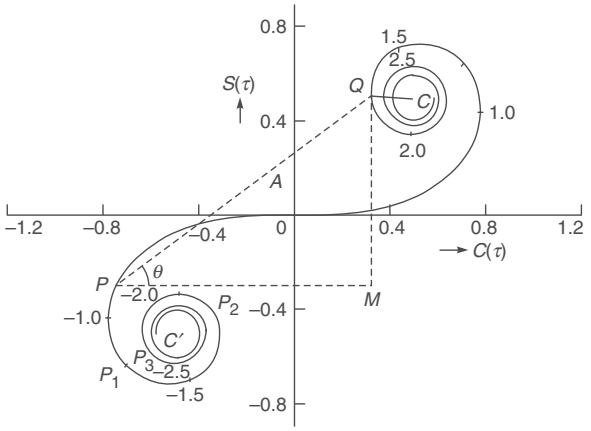


Fig. 20.11 The Cornu's spiral which is a parametric plot of $C(\tau)$ and $S(\tau)$.

We will now use Eq. (20.23) to calculate the diffraction pattern of a plane wave incident normally on a straight edge (see Fig. 20.12). Obviously, there will be no variation of intensity along the x-axis and, therefore, without any loss of generality, we may assume the coordinates of an arbitrary point P (on the screen) to be $(0, y)$, where the origin has been assumed to be on the edge of the geometrical shadow. If the x and y coordinates of an arbitrary point M on the plane of the straight edge are denoted by ξ and η, then

$$\begin{aligned} r = MP &= [\xi^2 + (\eta - y)^2 + d^2]^{1/2} \\ &= d\left[1 + \frac{\xi^2 + (\eta - y)^2}{d^2}\right]^{1/2} \\ &\approx d + \frac{\xi^2 + (\eta - y)^2}{2d} \end{aligned} \qquad \textbf{(20.49)}$$

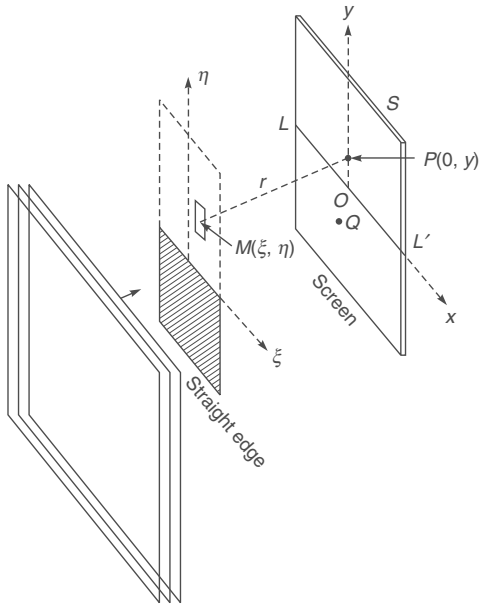


Fig. 20.12 Diffraction of a plane wave incident normally on a straight edge.

where d is the distance between the straight edge and the screen. On substituting the expression for r from Eq. (20.49) in Eq. (20.21), we obtain

$$u(P) \approx -\frac{i}{\lambda}\frac{A}{d}\int_{-\infty}^{\infty} d\xi \int_0^{\infty} d\eta \exp\left[ik\left\{d + \frac{\xi^2 + (\eta - y)^2}{2d}\right\}\right] \qquad \textbf{(20.50)}$$

where, in the denominator of the integrand, we have replaced r by its minimum value**, d. In order to express the above expression in terms of the Fresnel integrals, we introduce two dimensionless variables u and v such that

* Table adapted from Ref. 20.5; a more detailed table (with greater accuracy) has been given there.

** This is justified because in carrying out the integration, only a small region around the point $r = d$ contributes; the contribution due to far-off points is small because of the rapid oscillations of the exponential term in the integrand (see also footnote in Sec. 19.2).

$$\text{and} \quad \left.\begin{aligned} \frac{1}{2}\pi u^2 &= \frac{k}{2d}\xi^2 = \frac{\pi}{\lambda d}\xi^2 \\ \frac{1}{2}\pi v^2 &= \frac{k}{2d}(\eta - y)^2 = \frac{\pi}{\lambda d}(\eta - y)^2 \end{aligned}\right\}$$

Thus we may assume u and v to be *defined* by the following equations:

$$\text{and} \quad \left.\begin{aligned} u &= \sqrt{\frac{2}{\lambda d}}\,\xi \\ v &= \sqrt{\frac{2}{\lambda d}}\,(\eta - y) \end{aligned}\right\} \qquad \textbf{(20.51)}$$

With these substitutions, Eq. (20.51) becomes

$$u(P) = -\frac{i}{2}u_0 \int_{-\infty}^{+\infty} \exp\left(\frac{i\pi u^2}{2}\right) du \int_{v_0}^{\infty} \exp\left(\frac{i\pi v^2}{2}\right) dv \qquad \textbf{(20.52)}$$

where

$$v_0 = -\sqrt{\frac{2}{\lambda d}}\,y \qquad \textbf{(20.53)}$$

and

$$u_0 = A\,e^{ikd}$$

represents the field at the point P in the absence of the straight edge. In order to calculate the intensity distribution we use the Fresnel integrals; thus,

$$\int_{-\infty}^{+\infty} \exp\left[i\frac{\pi u^2}{2}\right] du = 2\,[C(\infty) + S(\infty)]$$

$$= (1 + i) \qquad \textbf{(20.54)}$$

Further,

$$\int_{v_0}^{\infty} \exp\left(\frac{i\pi v^2}{2}\right) dv = \left[\int_0^{\infty} \cos\left(\frac{\pi}{2}v^2\right) dv - \int_0^{v_0} \cos\left(\frac{\pi}{2}v^2\right) dv\right]$$

$$+ i\left[\int_0^{\infty} \sin\left(\frac{\pi}{2}v^2\right) dv - \int_0^{v_0} \sin\left(\frac{\pi}{2}v^2\right) dv\right]$$

$$= \left[\frac{1}{2} - C(v_0)\right] + i\left[\frac{1}{2} - S(v_0)\right] \qquad \textbf{(20.55)}$$

Substituting in Eq. (20.52), we obtain

$$u(P) = -\frac{i}{2}u_0(1+i)\left[\left\{\frac{1}{2} - C(v_0)\right\} + i\left\{\frac{1}{2} - S(v_0)\right\}\right]$$

$$= \frac{1-i}{2}u_0\left[\left\{\frac{1}{2} - C(v_0)\right\} + i\left\{\frac{1}{2} - S(v_0)\right\}\right] \qquad \textbf{(20.56)}$$

It is of interest to note that a large value of y corresponds to a point which is very far above the edge of the geometrical shadow. For such a point v_0 would tend to $-\infty$ [see Eq. (20.53)] and we would obtain

$$u(P) = \frac{1-i}{2}u_0\left[\left(\frac{1}{2} + \frac{1}{2}\right) + i\left(\frac{1}{2} + \frac{1}{2}\right)\right]$$

$$= u_0 \qquad \textbf{(20.57)}$$

Thus, as expected, the amplitude at such a point is the same as that in the absence of the edge. This also justifies the value of the constant given by Eq. (19.3). On the other hand, when the point P is deep inside the geometrical shadow (i.e., when $y \to -\infty$ and hence $v_0 \to \infty$), we obtain

$$C(v_0) = S(v_0) \to \tfrac{1}{2}$$

giving

$$u(P) \to 0$$

as it should indeed be. The intensity distribution corresponding to Eq. (20.56) would be given by

$$I(P) = \frac{1}{2}I_0\left[\left\{\frac{1}{2} - C(v_0)\right\}^2 + \left\{\frac{1}{2} - S(v_0)\right\}^2\right] \qquad \textbf{(20.58)}$$

If the point P is such that it lies on the edge of the geometrical shadow (i.e., on the line LL' (see Fig. 20.12) then $y = 0$ and hence $v_0 = 0$; thus

$$I(P) = \frac{1}{2}I_0\left[\frac{1}{4} + \frac{1}{4}\right] = \frac{1}{4}I_0 \qquad \textbf{(20.59)}$$

where we have used the fact that $C(0) = S(0) = 0$. Thus the intensity on the edge of the geometrical shadow is $1/4^{\text{th}}$ of the intensity that would have been in the absence of the edge. In order to determine the field at an arbitrary point P, we may use Table 20.1 to calculate the RHS of Eq. (20.58). The intensity variation is plotted in Fig. 20.16 from which one can make the following observations:

(i) Figure 20.13 represents a universal curve, i.e., for given values of λ and d, one simply has to calculate v_0 as the observation point moves along the y-axis. For example, the first three maxima occur at

$$\text{and} \quad \left.\begin{aligned} v_0 &\approx -1.22 \ \text{ with } \ I \approx 1.37\,I_0 \\ v_0 &\approx -2.34 \ \text{ with } \ I \approx 1.20\,I_0 \\ v_0 &\approx -3.08 \ \text{ with } \ I \approx 1.15\,I_0 \end{aligned}\right\} \text{ maxima}$$

Similarly, the first three minima occur at

$$\text{and} \quad \left.\begin{aligned} v_0 &\approx -1.87 \ \text{ with } \ I \approx 0.778\,I_0 \\ v_0 &\approx -2.74 \ \text{ with } \ I \approx 0.843\,I_0 \\ v_0 &\approx -3.39 \ \text{ with } \ I \approx 0.872\,I_0 \end{aligned}\right\} \text{ minima}$$

Thus, as we go inside the geometrical shadow, the intensity modulation decreases (see Fig. 20.13).

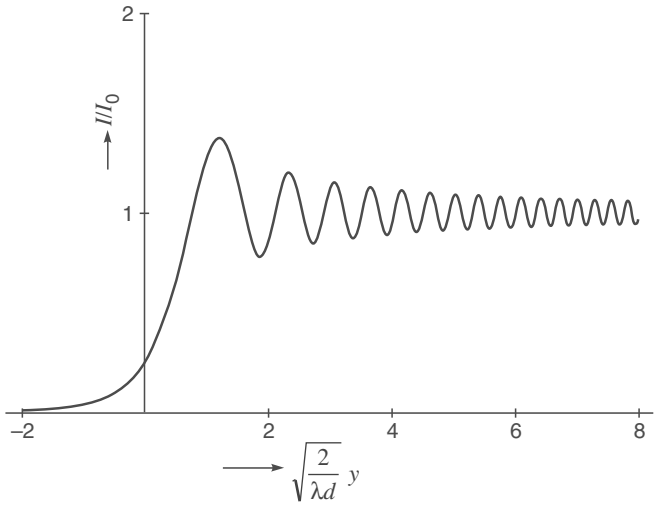


Fig. 20.13 The intensity variation corresponding to the straight edge diffraction pattern.

(ii) For a given experimental set-up, the determination of the positions of maxima and minima is quite straightforward. For example, for $\lambda = 6 \times 10^{-5}$ cm and $d = 120$ cm

$$y = -\sqrt{\frac{\lambda d}{2}}\, v_0 = -0.06\, v_0 \text{ cm}$$

Thus, the first three maxima will occur at

$$y \approx 0.732,\ 1.404 \quad \text{and} \quad 1.848 \text{ mm}$$

respectively. Similarly, the first three minima will occur at

$$y \approx 1.122,\ 1.644 \text{ and } 2.034 \text{ mm}$$

respectively.

(iii) As we go inside the geometrical shadow the intensity monotonically decreases to zero.

(iv) One could have also studied the intensity variation directly from the Cornu's spiral (see Fig. 20.11). This is due to the fact that associated with the Cornu's spiral, we have the following interesting property: let us write

$$[C(\tau_2) - C(\tau_1)] + i[S(\tau_2) - S(\tau_1)] \equiv A\, e^{i\theta} \qquad \textbf{(20.60)}$$

Thus,

$$[C(\tau_2) - C(\tau_1)] = A \cos\theta$$

and

$$[S(\tau_2) - S(\tau_1)] = A \sin\theta$$

Let the points P and Q on the Cornu's spiral (see Fig. 20.11) correspond to $\tau = \tau_1$ and $\tau = \tau_2$ respectively. It is obvious that

$$PM = [C(\tau_2) - C(\tau_1)] = A \cos\theta$$

and

$$QM = [S(\tau_2) - S(\tau_1)] = A \sin\theta$$

Thus the length of the line joining the points P and Q will be A and the angle that the line makes with the abscissa will be θ. In order to use the Cornu's spiral we rewrite Eq. (20.56):

$$u = \frac{1-i}{2}\, u_0 \left[\left\{ \frac{1}{2} - C(v_0) \right\} + i \left\{ \frac{1}{2} - S(v_0) \right\} \right]$$

Let us first consider a point of observation Q in the geometrical shadow region. Consequently v_0 will be positive. Let the point Q on the spiral (see Fig. 20.11) correspond to $\tau = v_0$. Since the point C in the curve corresponds to $\tau = \infty$, we have

$$\left\{ \frac{1}{2} - C(v_0) \right\} + i \left\{ \frac{1}{2} - S(v_0) \right\} = (QC)\, e^{i\psi}$$

where ψ is the angle that QC makes with the abscissa [see Eq. (20.60)]. Thus,

$$u(Q) = \frac{1-i}{2}\, (QC)\, e^{i\psi}\, u_0$$

or

$$I(Q) = \frac{1}{2}\, (QC)^2\, I_0 \qquad \textbf{(20.61)}$$

We can easily see that as the point of observation moves into the shadow region, the value of v_0 increases. Thus the point Q keeps on moving on the spiral towards the point C and the length QC decreases uniformly. Hence in the shadow region the intensity uniformly decreases to zero (see Figs. 20.13 and 20.14).

As we move away from the edge of the geometrical shadow to the illuminated region, the value of v_0 becomes negative and the corresponding point P (on the Cornu's spiral) lies in the third quadrant as shown in Fig. 20.11. The intensity is again given by

$$I(P) = \frac{1}{2}\, (PC)^2\, I_0$$

As the value of v_0 becomes more and more negative, the length PC keeps on increasing till the point P reaches P_1 which corresponds to $v_0 \approx -1.22$. The intensity at this point is maximum and the length $P_1C \approx 1.66$. Thus, the corresponding intensity is

$$I(P_1) \approx \frac{1}{2}\, (1.66)^2\, I_0 \approx 1.37\, I_0 \qquad \textbf{(20.62)}$$

As the value of v_0 becomes further negative the length PC starts decreasing till it reaches the point P_2. Thus,

the intensity keeps on oscillating with decreasing amplitude about I_0 as we move more and more into the illuminated region (see Figs. 20.13 and 20.14).

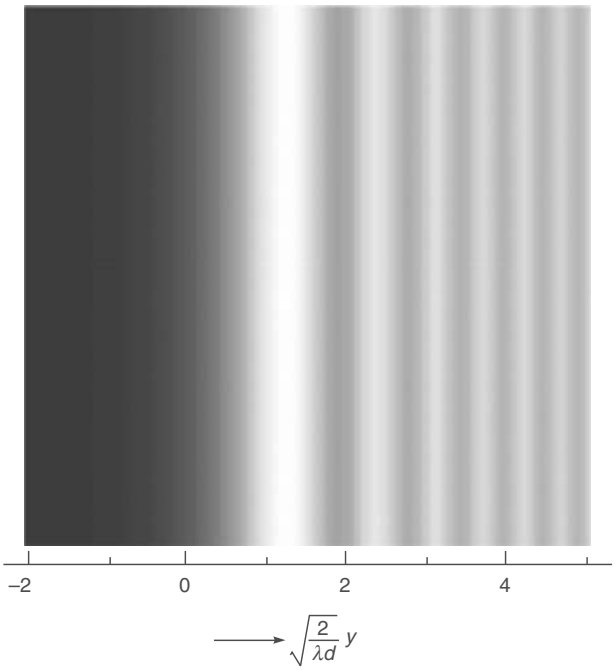


Fig. 20.14 Computer generated intensity distribution corresponding to the straight edge diffraction pattern

20.7 DIFFRACTION OF A PLANE WAVE BY A LONG NARROW SLIT AND TRANSITION TO THE FRAUNHOFER REGION LO 5

We next consider a plane wave incident normally on a long narrow slit (of width b) as shown in Fig. 20.15. We wish to calculate the intensity distribution at an arbitrary point P on the screen SS'. The lines LL' and MM' represent the edges of the geometrical shadow. Once again, there will be no variation of the intensity along the x-axis and we may (without any loss of generality) assume the coordinates of the point P to be $(0, y)$. The field at the point P will again be given by Eq. (20.50) except that the limits of the η-integral will be $-b/2$ and $+b/2$ (we are assuming the origin to be at the centre of the slit)

$$u(P) = -\frac{i\,A}{\lambda d}\int_{-\infty}^{\infty} d\xi \int_{-b/2}^{+b/2} d\eta \exp\left(ik\left\{d + \frac{\xi^2 + (\eta - y)^2}{2d}\right\}\right)$$

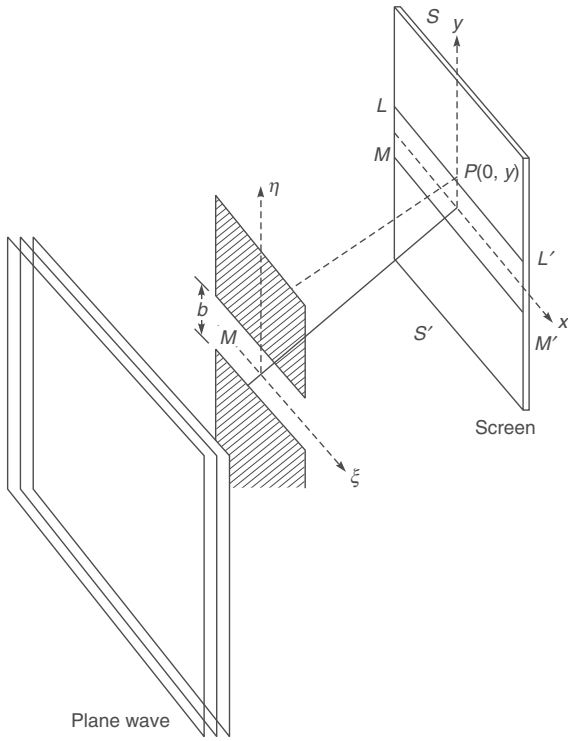


Fig. 20.15 Diffraction of a plane wave incident normally on a long narrow slit.

Carrying out manipulations similar to that in the previous section, we obtain

$$u(P) = -\frac{i}{2}u_0 \int_{-\infty}^{\infty} \exp\left(\frac{i\pi u^2}{2}\right) du \int_{-(v_2+v_1)}^{-(v_2-v_1)} \exp\left(\frac{i\pi v^2}{2}\right) dv$$

where

$$u = \sqrt{\frac{2}{\lambda d}}\,\xi, \qquad v = \sqrt{\frac{2}{\lambda d}}\,(\eta - y)$$

and

$$v_1 = \sqrt{\frac{2}{\lambda d}}\,\frac{b}{2}, \qquad v_2 = \sqrt{\frac{2}{\lambda d}}\,y$$

Using Eq. (20.46) we obtain

$$u(P) = -\frac{i}{2}u_0(1+i)\times$$

$$\left\{\int_0^{-(v_2-v_1)} \left[\cos\left(\frac{\pi}{2}v^2\right) + i\sin\left(\frac{\pi}{2}v^2\right)\right] dv\right.$$

$$\left. - \int_0^{-(v_2+v_1)} \left[\cos\left(\frac{\pi}{2}v^2\right) + i\sin\left(\frac{\pi}{2}v^2\right)\right] dv\right\}$$

or

$$u(P) = \frac{(1-i)}{2} u_0 [\{C(v_2 + v_1) - C(v_2 - v_1)\} + i \{S(v_2 + v_1) - S(v_2 - v_1)\}] \quad \textbf{(20.63)}$$

where we have used Eq. (20.44). Thus, the intensity distribution would be

$$I(P) = \frac{1}{2} I_0 [\{C(v_2 + v_1) - C(v_2 - v_1)\}^2 + \{S(v_2 + v_1) - S(v_2 - v_1)\}^2] \quad \textbf{(20.64)}$$

For a given system λ, d and b are known which determine v_1; e.g., for $\lambda = 5 \times 10^{-5}$ cm, $d = 100$ cm and $b = 0.1$ cm, one obtains $v_1 = 2.0$; further, as y varies on the screen, the quantity v_2 also changes. In Figs. 20.16, 20.17, 20.18 and 20.19 we have plotted the intensity variation as a function of v_2 for $v_1 = 0.5$, 1.0, 1.5 and 5.0, respectively. One can see that for a large value of v_1 (i.e., when the slit width is very large) the diffraction pattern is similar to that produced by two straight edges. This is indeed what we should have also expected. On the other hand, for small values of v_1 (i.e., when the observation screen is far away from the aperture) the diffraction pattern is essentially of the Fraunhofer type. In order to show this explicitly we notice that

$$v_2 = \sqrt{\frac{2}{\lambda d}}\, y = \sqrt{\frac{2d}{\lambda}} \frac{y}{d} \approx \sqrt{\frac{2d}{\lambda}}\, \theta \quad \textbf{(20.65)}$$

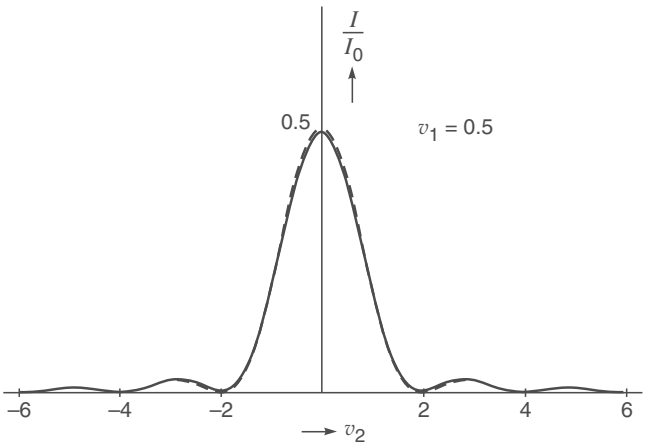


Fig. 20.16 The intensity distribution produced by diffraction of a plane wave by a long narrow slit corresponding to $v_1 = 0.5$. The dashed curves correspond to Eq. (20.66).

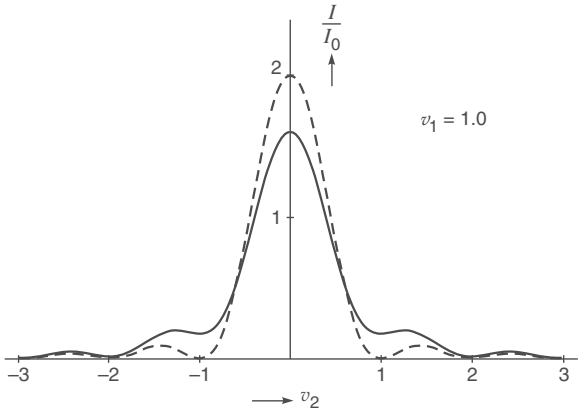


Fig. 20.17 The intensity distribution produced by diffraction of a plane wave by a long narrow slit corresponding to $v_1 = 1.0$. The dashed curves correspond to Eq. (20.66).

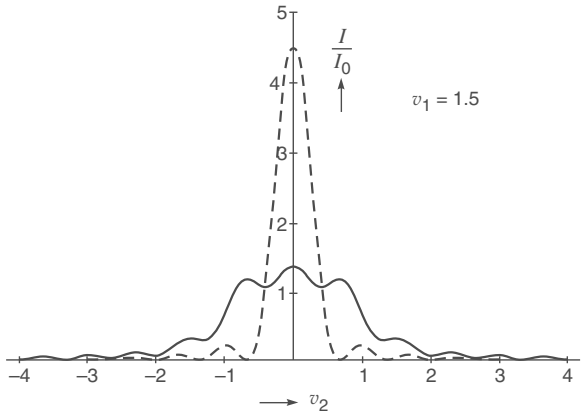


Fig. 20.18 The intensity distribution produced by diffraction of a plane wave by a long narrow slit corresponding to $v_1 = 1.5$. The dashed curves correspond to Eq. (20.66).

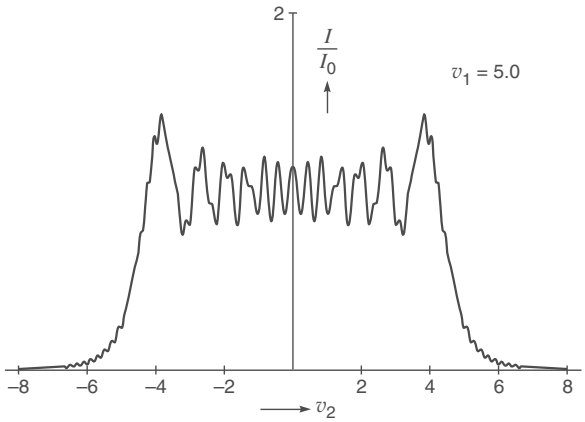


Fig. 20.19 The intensity distribution produced by diffraction of a plane wave by a long narrow slit corresponding to $v_1 = 5.0$.

where θ represents the angle of diffraction (see Fig. 20.20). Clearly, in the Fraunhofer region since d is very large, the value of v_2 will also be very large and thus we must look for expressions of the Fresnel integrals in the limit of $v \to \infty$. Now, we may write

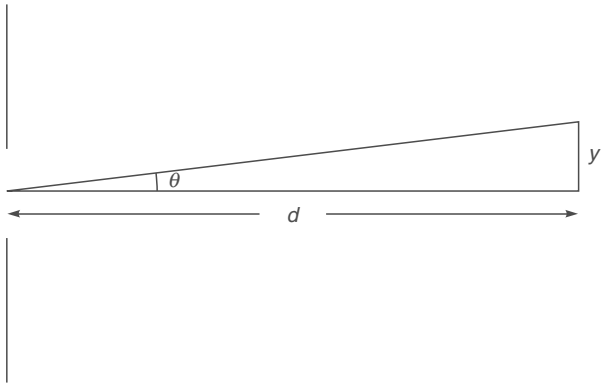


Fig. 20.20 In the Fraunhofer region, d is very large.

$$C(v) = \int_0^v \cos\frac{\pi}{2}v^2\,dv$$

$$= \int_0^\infty \cos\frac{\pi}{2}v^2\,dv - \int_v^\infty \cos\frac{\pi}{2}v^2\,dv$$

$$= \frac{1}{2} - \int_v^\infty \frac{1}{\pi v}\cos\left(\frac{\pi}{2}v^2\right)\pi v\,dv$$

$$= \frac{1}{2} - \frac{1}{\pi v}\sin\left(\frac{\pi}{2}v^2\right)\Big|_v^\infty + \int_v^\infty \frac{1}{\pi v^2}\sin\left(\frac{\pi}{2}v^2\right)dv$$

$$\cong \frac{1}{2} + \frac{1}{\pi v}\sin\left(\frac{\pi}{2}v^2\right)$$

where we have neglected terms which would be of order $1/v^3$. Similarly,

$$S(v) = \frac{1}{2} - \frac{1}{\pi v}\cos\left(\frac{\pi}{2}v^2\right)$$

Since v_2 is large and v_1 is small, we have

$$C(v_2 + v_1) - C(v_2 - v_1) \approx \left[\frac{1}{2} + \frac{1}{\pi v_2}\sin\frac{\pi}{2}(v_2 + v_1)^2\right]$$

$$- \left[\frac{1}{2} + \frac{1}{\pi v_2}\sin\frac{\pi}{2}(v_2 - v_1)^2\right]$$

$$\cong \frac{2}{\pi v_2}\cos\frac{\pi}{2}(v_2^2 + v_1^2)\sin\pi v_1 v_2$$

Similarly,

$$S(v_2 + v_1) - S(v_2 - v_1) \approx \frac{2}{\pi v_2}\sin\frac{\pi}{2}(v_2^2 + v_1^2)\sin(\pi v_1 v_2)$$

Thus, in the Fraunhofer limit, Eq. (20.64) becomes

$$I(P) = \frac{1}{2}I_0\left[\frac{4}{\pi^2 v_2^2}\sin^2(\pi v_1 v_2)\right]$$

$$= I_{00}\frac{\sin^2\beta}{\beta^2} \qquad \textbf{(20.66)}$$

where

$$I_{00} = 2\,I_0\,v_1^2$$

and

$$\beta = \pi v_1 v_2 = \frac{\pi}{\lambda d}b\,y \approx \frac{\pi b}{\lambda}\theta \qquad \textbf{(20.67)}$$

and

$$\theta \approx \frac{y}{d} \qquad \textbf{(20.68)}$$

represents the diffraction angle. Equation (20.66) shows that the intensity distribution is indeed of the Fraunhofer type (see Sec. 18.2). In Figs. 20.16 – 20.19 the dashed curves correspond to Eq. (20.66) and one can see that the intensity distribution is almost of the Fraunhofer type for $v_1 \leq 0.5$.

Summary

- The underlying principle in the theory of diffraction is the Huygens—Fresnel principle according to which: *Each point on a wavefront is a source of secondary disturbance and the secondary wavelets emanating from different points mutually interfere*.
- For a plane wave incident normally on a circular aperture of radius a, the intensity variation on an axial point P is given by

  $$I = I_0\sin^2\frac{p\pi}{2}$$

  where

  $$p \approx \frac{a^2}{\lambda d}$$

  λ is the wave length and d is the distance of the point P from the center of the circular aperture. The quantity p is known as the Fresnel number of the aperture. When p = 1,3,5,7, … we have maximum intensity and the circular aperture will contain (with respect to the point P) odd number of Fresnel half period zones and when p = 2,4,6,8 … we have minimum intensity and the circular aperture will contain even number of half-period zones.
- If instead of the circular aperture we have opaque disc, then we would always obtain a bright spot on the axis behind the disc; this is called the 'Poisson spot'.

- For a Gaussian beam (whose phase front is plane at $z = 0$), the variation of the spot size is given by

$$w(z) \approx w_0\left[1+\frac{\lambda^2 z^2}{\pi^2 w_0^4}\right]^{1/2}$$

where w_0 is the spot size at $z = 0$. For large values of z:

$$w(z) \approx \frac{\lambda z}{\pi w_0}$$

which shows that the width increases linearly with z. We define the diffraction angle as

$$\tan\theta = \frac{w(z)}{z} \approx \frac{\lambda}{\pi w_0}$$

showing that the rate of increase in width is proportional to the wavelength and inversely proportional to the initial width of the beam; this is characteristic of diffraction. The corresponding radius of curvature of the wavefront is given by

$$R(z) \approx z\left[1+\frac{\pi^2 w_0^4}{\lambda^2 z^2}\right]$$

- For a plane wave incident normally on a straight edge, the intensity variation on a screen (at a distance d from the straight edge) is given by

$$I = \frac{1}{2}I_0\left[\left\{\frac{1}{2}-C(v_0)\right\}^2+\left\{\frac{1}{2}-S(v_0)\right\}^2\right]$$

where I_0 is the intensity in the absence of the straight edge,

$$v_0 = -\sqrt{\frac{2}{\lambda d}}\,y$$

y being the distance from the edge of the geometrical shadow and

$$C(x) = \int_0^x \cos\left(\tfrac{1}{2}\pi u^2\right)du$$

and

$$S(x) = \int_0^x \sin\left(\tfrac{1}{2}\pi u^2\right)du$$

are known as Fresnel integrals. The intensity monotonically goes to zero as we go deep inside the geometrical shadow. As we move away from the edge of the geometrical shadow to the illuminated region, one obtains maxima at $v_0 \approx -1.22$ ($I \approx 1.37\ I_0$), -2.34 ($I \approx 1.20\ I_0$), -3.08 ($I \approx 1.15\ I_0$) … and minima at $v_0 \approx -1.87$ ($I \approx 0.78\ I_0$), -2.74 ($I \approx 0.84\ I_0$), -3.39 ($I \approx 0.87\ I_0$), ...

- For a plane wave incident normally on a long narrow slit of width b, the intensity variation on a screen (at a distance d from the slit) is given by

$$I = \frac{1}{2}\,I_0\,[\{C(v_2+v_1)-C(v_2-v_1)\}^2 + \{S(v_2+v_1)-S(v_2-v_1)\}^2]$$

where

$$v_1 = \sqrt{\frac{2}{\lambda d}}\,\frac{b}{2};\ v_2 = \sqrt{\frac{2}{\lambda d}}\,y$$

and y is the distance from the midpoint of the edges of the geometrical shadow. As v_1 becomes large, we obtain the intensity distribution corresponding to two straight edges and for $v_1 \to 0$ we get the Fraunhofer diffraction pattern.

Problems

20.1 Consider a plane wave of wavelength 6×10^{-5} cm incident normally on a circular aperture of radius 0.01 cm. Calculate the positions of the brightest and the darkest points on the axis.

[**Ans:** $d \approx 1.67$ cm, 0.56 cm, 0.33 cm, … (Maxima); $d \approx 0.83$ cm, 0.42 cm, …(Minima)]

20.2 What would happen if the circular aperture in Problem 20.1 is replaced by a circular disc of the same radius?

20.3 **(a)** A plane wave ($\lambda = 6 \times 10^{-5}$ cm) is incident normally on a circular aperture of radius a.

(i) Assume $a = 1$ mm. Calculate the values of z (on the axis) for which maximum intensity will occur. Plot the intensity as a function of z and interpret physically.

(ii) Assume $z = 50$ cm. Calculate the values of a for which minimum intensity will occur on the axial point. Plot the intensity variation as a function of a and interpret physically.

(b) Repeat the calculations for $\lambda = 5 \times 10^{-5}$ cm and discuss chromatic aberration of a zone plate.

[**Ans:** (i) (a) $z \approx 166.7$ cm, 55.6 cm, 33.3 cm,… (maxima); (b) Minimum intensity will occur when $a \approx 0.0775$ cm, 0.110 cm, 0.134 cm, …]

20.4 Consider a circular aperture of diameter 2 mm illuminated by a plane wave. The most intense point on the axis is at a distance of 200 cm from the aperture. Calculate the wavelength.

[**Ans.** 5×10^{-5} cm]

20.5 If a zone-plate has to have a principle focal length of 50 cm corresponding to $\lambda = 6 \times 10^{-5}$ cm, obtain an expression for the radii of different zones. What would be its principle focal length for $\lambda = 5 \times 10^{-5}$ cm?

[$(\sqrt{0.3n})$ mm, 60 cm]

20.6 In a zone-plate, the second, fourth, sixth…zones are blackened; what would happen if instead the 1st, 3rd, 5th, etc., zones were blackened?

20.7 **(a)** A plane wave is incident normally on a straight edge (see Fig. 20.21). Show that the field at an arbitrary point P is given by

$$u(P) = \frac{1-i}{2}u_0\left[\left\{\frac{1}{2}-C(v_0)\right\}+i\left\{\frac{1}{2}-S(v_0)\right\}\right]$$

where $v_0 = -\sqrt{\frac{2}{\lambda d}}\,y$.

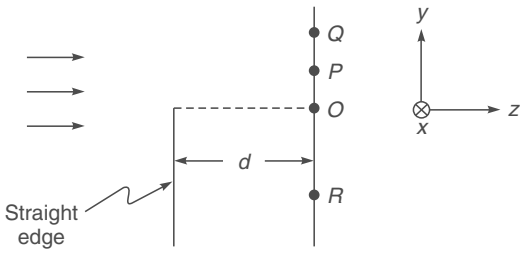


Fig. 20.21

(b) Assume $\lambda_0 = 5000$ Å and $d = 100$ cm. Using Table 20.1, write approximately the values of I/I_0 at the points O, P ($y = 0.5$ mm), Q ($y = 1$ mm) and R ($y = -1$ mm) where O is at the edge of the geometrical shadow.

[**Ans:** (b) $I/I_0 \approx 1.26, 0.24, 0.01$]

20.8 Consider a straight edge being illuminated by a parallel beam of light with $\lambda = 6 \times 10^{-5}$ cm. Calculate the positions of the first two maxima and minima on a screen at a distance of 50 cm from the edge.

[**Ans:** the first two maxima occur at $y \approx 0.0473$ cm and 0.0906 cm. The first two minima occur at $y \approx 0.0724$ cm and 0.1061 cm.]

20.9 In a straight edge diffraction pattern, one observes that the most intense maximum occurs at a distance of 1 mm from the edge of the geometrical shadow. Calculate the wavelength of light, if the distance between the screen and the straight edge is 300 cm.

[**Ans.** ≈ 4480 Å]

20.10 In a straight edge diffraction pattern, if the wavelength of the light used is 6000 Å and if the distance between the screen and the straight edge is 100 cm, calculate the distance between the most intense maximum and the next maximum. Find approximately the distance in centimeters inside the geometrical shadow where $I/I_0 = 0.1$.

[**Ans:** $y \approx 0.027$ cm]

20.11 Consider a plane wave falling normally on a narrow slit of width 0.5 mm. If the wavelength of light is 6×10^{-5} cm, calculate the distance between the slit and the screen so that the value of v_1 would be 0.5, 1.0, 1.5 and 5.0 (see Figs. 20.16–20.19). Discuss the transition to the Fraunhofer region.

20.12 Consider the Fresnel diffraction pattern produced by a plane wave incident normally on a slit of width b. Assume $\lambda = 5 \times 10^{-5}$ cm, $d = 100$ cm. Using Table 20.1, approximately calculate the intensity values (for $b = 0.1$ cm) at $y = 0, \pm 0.05$ cm, ± 0.1 cm. Repeat the analysis for $b = 5$ cm.

[**Ans:** At $y = 0$, $I/I_0 \approx 1.60$; at $y = \pm 0.05$ cm, $I/I_0 \approx 0.356$; at $y = \pm 0.01$ cm, $I/I_0 \approx 0.01685$].

20.13 **(a)** The output of a He–Ne laser ($\lambda = 6328$ Å) can be assumed to be Gaussian with plane phase front. For $w_0 = 1$ mm and $w_0 = 0.2$ mm, calculate the beam diameter at $z = 20$ m.

(b) Repeat the calculation for $\lambda = 5000$ Å and interpret the results physically.

[**Ans:** (a) 0.83 cm and 4.0 cm]

20.14 A Gaussian beam is coming out of a laser. Assume $\lambda = 6000$ Å and that at $z = 0$, the beam width is 1 mm and the phase front is plane. After traversing 10 m through vacuum what will be **(a)** the beam width and **(b)** the radius of curvature of the phase front.

[**Ans:** $2w \approx 0.77$ cm; $R(z) \approx 1017$ cm]

20.15 A plane wave of intensity I_0 is incident normally on a circular aperture as shown in Fig. 20.22. What will be the intensity on the axial point P? [**Ans:** $3I_0$]

[**Hint:** You may use Eq. (20.27)]

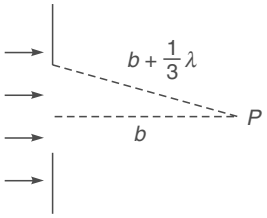


Fig. 20.22

20.16 Show that a phase variation of the type

$$\exp\left[ikz + \frac{ik(x^2 + y^2)}{2R(z)}\right]$$

represents a diverging spherical wave of radius R.

20.17 The output of a semiconductor laser can be approximately described by a Gaussian function with two different widths along the transverse (w_T) and lateral (w_L) directions as

$$\psi(x, y) = A \exp\left(-\frac{x^2}{w_L^2} - \frac{y^2}{w_T^2}\right)$$

where x and y represent axes parallel and perpendicular to the junction plane. Typically $w_T \approx 0.5$ μm and $w_L = 2$ μm. Discuss the far field of this beam (see Fig. 20.23).

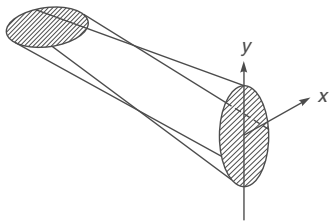


Fig. 20.23

Chapter Twenty One

HOLOGRAPHY*

The electron microscope was to produce the interference figure between the object beam and the coherent background, that is to say the non-diffracted part of the illuminating beam. This interference pattern I called a ***hologram****, from the Greek word* ***holos*** *– the whole, because it contained the whole information. The hologram was then reconstructed with light, in an optical system which corrected the aberrations of the electron optics.*

—Dennis Gabor in his Nobel lecture** December 11, 1971

Learning Objectives

After reading this chapter, the reader should be able to:

LO 1: explain the phenomenon of holography.
LO 2: describe the requirements needed for making a hologram.
LO 3: apply holography in diverse fields.

Important Milestones

1948 *Dennis Gabor discovered the principle of holography.*
1960 *The first successful operation of a laser device by Theodore Maiman.*
1962 *Off-axis technique of holography by Leith and Upatnieks.*
1962 *Denisyuk suggested the idea of three-dimensional holograms based on thick photoemulsion layers. His holograms can be reconstructed in ordinary sun light. These holograms are called Lippmann – Bragg holograms.*
1964 *Leith and Upatnieks pointed out that a multi-color image can be produced by a hologram recorded with three suitably chosen wavelengths.*
1969 *Benton invented 'Rainbow Holography' for display of holograms in white light. This was a vital step to make holography suitable for display applications.*

21.1 INTRODUCTION

A photograph represents a two-dimensional recording of a three-dimensional scene. What is recorded is the intensity distribution that prevailed at the plane of the photograph when it was exposed. The light sensitive medium is sensitive only to the intensity variations and hence while recording a photograph, the phase distribution which prevailed at the plane of the photograph is lost. Since only the intensity pattern has been recorded, the three-dimensional character (e.g., parallax) of the object scene is lost. Thus, one cannot change the perspective of the image in the photograph by

* A portion of this chapter is based on the unpublished lecture notes of Professor K. Thyagarajan.

** Dennis Gabor received the 1971 Nobel Prize in Physics for discovering the *principles of holography*; the original paper of Gabor appeared in 1948 [see Ref. 21.1]. Gabor's Nobel lecture entitled *Holography, 1948–1971* is non-mathematical and full of beautiful illustrations; it is reprinted in Ref. 22.2

viewing it from a different angle or one cannot refocus any unfocussed part of the image in the photograph. Holography is a method invented by Dennis Gabor in 1947, in which one not only records the amplitude but also the phase of the light wave; this is done by using interferometric techniques. Because of this, the image produced by the technique of holography has a true three-dimensional form. Thus, as with the object, one can change one's position and view a different perspective of the image or one can focus at different distances. The capability to produce images as true as the object itself is what is responsible for the wide popularity gained by holography.

The basic technique in holography is the following: In the recording of the hologram, one superimposes on the object wave another wave called the reference wave and the photographic plate is made to record the resulting interference pattern (see Fig. 21.1). The reference wave is usually a plane

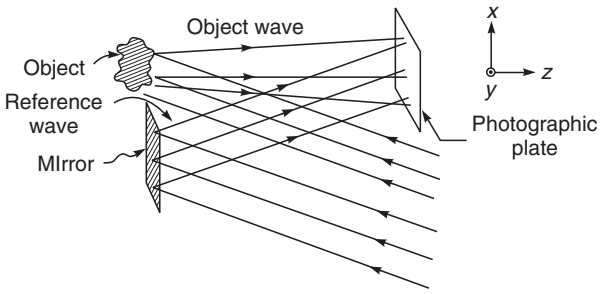


Fig. 21.1 Recording of a hologram.

wave. This recorded interference pattern forms the hologram and (as will be shown) contains information not only about the amplitude but also about the phase of the object wave. Unlike a photograph, a hologram has little resemblance with the object; in fact, information about the object is coded into the hologram. To view the image, we again illuminate the hologram with another wave, called the **reconstruction wave** (which in most cases is identical to the reference wave used during the formation of the hologram); this process is termed as **reconstruction** (see Fig. 21.2). **The reconstruction process** leads, in general, to a virtual and a real image of the object scene. The virtual image has all the characteristics of the object like parallax, etc. Thus, one can move the position of the eye and look behind the objects or one can focus at different distances. The real image can be photographed without the aid of lenses just by placing a light sensitive medium at the position where the real image is formed. Figures 21.3(a), (b) and (c) represent the object, its hologram and the reconstructed image, respectively.

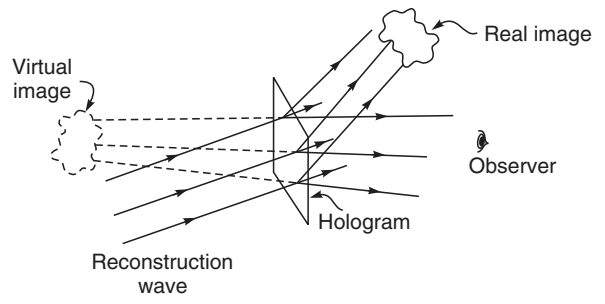


Fig. 21.2 Reconstruction process.

21.2 BASIC THEORY LO 1

If the object is a point scatterer, then the object wave would just be $\frac{A}{r}\cos(kr - \omega t + \phi)$ where r represents the distance of the point of observation from the point scatterer and A represents a constant; $k = 2\pi/\lambda$. Any general object can be thought of as being made up of a large number of points and the composite wave reflected by the object would be vectorial sum of these. The fundamental problem in holography is the recording of this object wave, in particular, the phase distribution associated with it.

Let us consider the recording process. Let

$$O(x, y) = a(x, y)\cos[\phi(x, y) - \omega t] \qquad \textbf{(21.1)}$$

represents the object wave (which, as mentioned earlier, is due to the superposition of waves from point scatterers on the object) in the plane of the photographic plate which is assumed to be $z = 0$ (see Fig. 21.1). We consider a plane reference wave and assume, for simplicity, that it is pro-pagating in the x–z-plane inclined at an angle θ with the z-direction (see Fig. 21.1). Thus, the field associated with this plane wave would be given by

$$\begin{aligned} r(x, y, z) &= A\cos[\mathbf{k}\,.\,\mathbf{r} - \omega t] \\ &= A\cos(kx\sin\theta + kz\cos\theta - \omega t) \end{aligned} \qquad \textbf{(21.2)}$$

If $r(x, y)$ represents the field at the plane $z = 0$ due to this reference wave, then one can see that

$$\begin{aligned} r(x, y) &= A\cos[kx\sin\theta - \omega t] \\ &= A\cos[2\pi\alpha x - \omega t] \end{aligned} \qquad \textbf{(21.3)}$$

where $\alpha = \sin\theta/\lambda$ is the spatial frequency (see Sec. 19.9). The above equation represents the field due to a plane wave inclined at an angle q with the z-axis and as can be seen the phase varies linearly with x. Notice that there is no y-dependence because the plane wave has been assumed to have its propagation vector in the x–z-plane. Thus, the total field at the photographic plate (which is coincident with the plane $z = 0$) would be given by

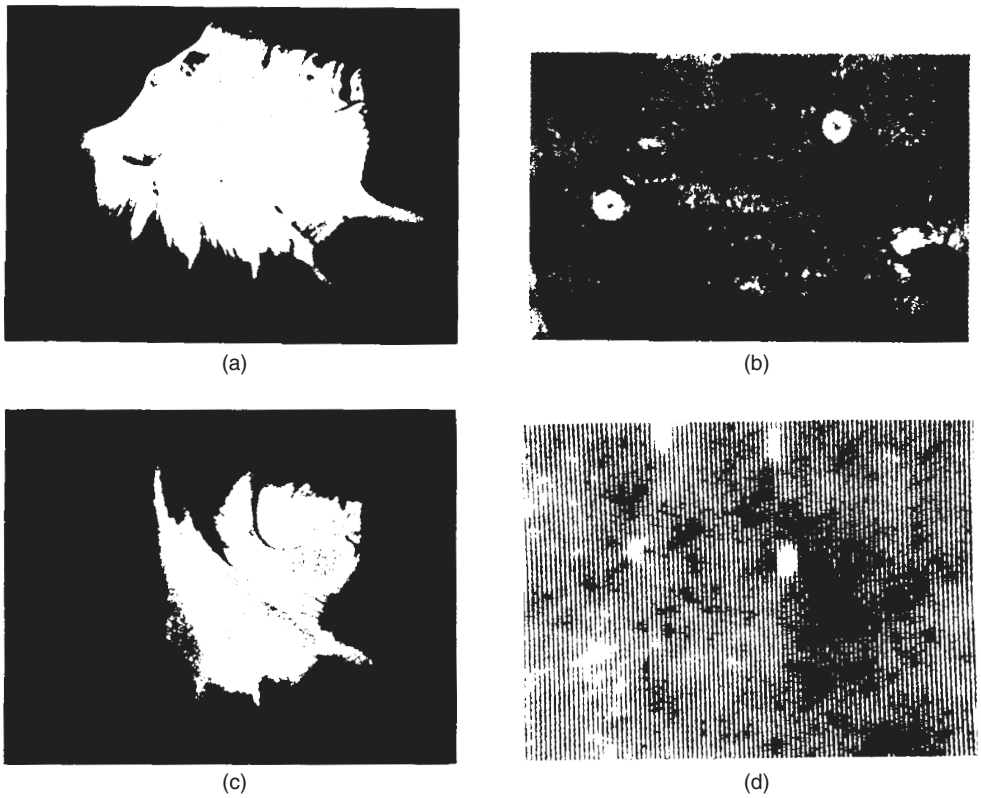


Fig. 21.3 (a) An ordinary photograph of an object. (b). The hologram of the object produced by a method similar to the one as shown in Fig. 21.1, (c) The reconstructed image as seen by an observer, (d) A magnified view of a small portion of the hologram shown in (b). [*Photographs courtesy: Professor R. S. Sirohi*].

$$u(x, y, t) = a(x, y) \cos [\phi(x, y) - \omega t] + A \cos [2\pi\alpha x - \omega t] \quad \textbf{(21.4)}$$

The photographic plate responds only to the intensity which would be proportional to the time average of $[u(x, y, t)]^2$. Thus, the intensity pattern recorded by the photographic plate would be

$$\begin{aligned} I(x, y) &= \langle u^2 (x, y, t) \rangle \\ &= \langle [a(x, y) \cos \{\phi (x, y) - \omega t\} \\ &\quad + A \cos(2\pi\alpha x - \omega t)]^2 \rangle \end{aligned} \quad \textbf{(21.5)}$$

where the angular brackets denote time averaging (see Sec. 17.3). Thus,

$$\begin{aligned} I(x, y) = a^2(x, y) \langle \cos^2 \{\phi(x, y) - \omega t\} \rangle \\ + A^2 \langle \cos^2 (2\pi \alpha x - \omega t) \rangle \\ + 2a(x, y) A \langle \cos \{\phi(x, y) - \omega t\} \cos (2\pi \alpha x - \omega t) \rangle \end{aligned} \quad \textbf{(21.6)}$$

Since

$$\langle \cos^2 [\phi(x, y) - \omega t] \rangle = \frac{1}{2} = \langle \cos^2 (2\pi\alpha x - \omega t) \rangle \quad \textbf{(21.7)}$$

and

$$\begin{aligned} &\langle \cos [\phi(x, y)) - \omega t] \cos (2\pi \alpha x - \omega t) \rangle \\ &\quad = \frac{1}{2} \langle \cos [\phi(x, y) + 2\pi \alpha x - 2 \omega t] \rangle + \\ &\qquad \frac{1}{2} \langle \cos [\phi(x, y) - 2\pi \alpha x] \rangle \\ &\quad = \frac{1}{2} \cos [\phi (x, y) - 2\pi\alpha x] \end{aligned} \quad \textbf{(21.8)}$$

Eq. (21.6) becomes

$$\begin{aligned} I (x, y) = \frac{1}{2} a^2(x, y) + \frac{1}{2} A^2 \\ + A a(x, y) \cos [\phi(x, y) - 2 \pi \alpha x] \end{aligned} \quad \textbf{(21.9)}$$

From the above relation, it is obvious that the phase information of the object wave, which is contained in ϕ (x, y), is recorded in the intensity pattern.

When the photographic plate (which has recorded the above intensity pattern) is developed, one obtains a hologram [see Figs. 21.3(b) and (d)]. The transmittance of the hologram, i.e., the ratio of the transmitted field to the incident field, depends on $I(x, y)$. By a suitable developing process one can obtain a condition under which the amplitude transmittance would be linearly related to $I(x, y)$. Thus, in such a case if $R(x, y)$ represents the field of the reconstruction wave at the hologram plane, then the transmitted field would be given by

$$\begin{aligned} v(x,y) &= K R(x,y)\, I(x,y) \\ &= K\left[\frac{1}{2}a^2(x,y) + \frac{1}{2}A^2\right] R(x,y) \\ &\quad + K A\, a(x,y)\, R(x,y) \cos\left[\phi(x,y) - 2\pi\alpha x\right] \end{aligned} \tag{21.10}$$

where K is a constant. We consider the case when the reconstruction wave is identical to the reference wave $r(x, y)$ (see Fig. 21.2). In such a case we would obtain (omitting the constant K)

$$\begin{aligned} v(x,y) &= \left[\frac{1}{2}a^2(x,y) + \frac{1}{2}A^2\right] A\cos(2\pi\alpha x - \omega t) \\ &\quad + A^2\, a(x,y)\cos[2\pi\alpha x - \omega t]\cos[\phi(x,y) - 2\pi\alpha x] \\ &= \left[\frac{1}{2}a^2(x,y) + \frac{1}{2}A^2\right] A\cos(2\pi\alpha x - \omega t) \\ &\quad + \frac{1}{2}A^2\, a(x,y)\cos[\phi(x,y) - \omega t] \\ &\quad + \frac{1}{2}A^2\, a(x,y)\cos[4\pi\alpha x - \phi(x,y) - \omega t] \end{aligned} \tag{21.11}$$

Equation (21.11) gives the transmitted field in the plane $z = 0$. We consider each of the three terms separately. The first term is nothing but the reconstruction wave itself whose amplitude is modulated due to the presence of the term $a^2(x, y)$. This part of the total field is traveling in the direction of the reconstructed wave. The second term is identical (within a constant term) to the RHS of Eq. (21.1) and hence represents the original object wave; this gives rise to a virtual image. Thus, the effect of viewing this wave is the same as viewing the object itself. The reconstructed object wave is traveling in the same direction as the original object wave.

To study the last term we first observe that in addition to the term $4\pi\alpha x$, the phase term $\phi(x, y)$ carries a negative sign. The negative sign represents the fact that the wave has a curvature opposite to that of the object wave. Thus, if the object wave is a diverging spherical wave then the last term represents a converging spherical wave. Thus, in contrast to the second term, this wave forms a real image of the object which can be photographed by simply placing a film (see Fig. 21.2).

To determine the effect of the term $4\pi\alpha x$, we consider the case when the object wave is also a plane wave traveling along the z-axis. For such a wave $\phi(x, y) = 0$ and the last term would represent a plane wave propagating along a direction $\theta' = \sin^{-1}(2 \sin \theta)$. Thus the effect of the term $4\pi\alpha x$ is to rotate the direction of the wave. Hence the last term on the RHS of Eq. (21.11) represents the conjugate of the object wave propagating along a direction different from that of the reconstruction wave and the object wave, which forms a real image of the object. Since the waves represented by the three terms are propagating along different directions they separate after traversing a distance and enable the observer to view the virtual image without any disturbance.

A very interesting property possessed by holograms is that even if the hologram is broken up into different fragments, each separate fragment is capable of producing a complete virtual image of the object.* This property can be understood from the fact that for a diffusely reflecting object, each point of the object illuminates the complete hologram and consequently each point in the hologram receives waves from the complete object. But the resolution in the image decreases as the size of the fragment decreases. For non-diffusely reflecting objects or for transparencies, one makes use of an additional diffusing screen through which the object is illuminated.

Example 21.1 As an explicit example of the formation and reconstruction of a hologram, we consider the simple case when both the object wave and the reference wave are plane waves [see Fig. 21.4(a)]—a plane object wave corresponds to a single object point lying far away from the hologram. (a) Show that for such a case, the hologram consists of a series of Young's interference fringes having an intensity distribution of the $\cos^2$ type. [(see also Fig. 14.11) (b)]. If we reconstruct the hologram with another plane wave [see Fig. 21.4(b)], then show that the transmitted light consists of a zero-order plane wave and two first-order plane waves; the two first-order waves correspond to the primary and conjugate waves.

* This property of a hologram exists only when the object is a diffuse scatterer such that the wave from each scattering point of the object reaches all parts of the hologram plate. There are cases where this does not hold good; for example, when a hologram of a transparency is to be recorded.

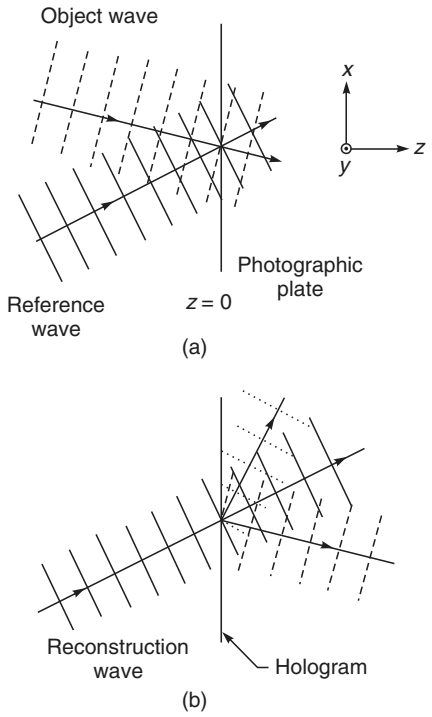


Fig. 21.4 (a) Formation of a hologram, when both the object wave and the reference wave are plane waves. (b) Reconstruction of the hologram with another plane wave.

Solution: (a) Consider a plane wave with its propagation vector lying in the x–z-plane and making an angle θ_1 with the z-axis. For such a wave, the field is of the form

$$A_1 \cos\ [kx \sin \theta_1 + kz \cos \theta_1 - \omega t]$$

If the photographic film is assumed to coincide with the plane $z = 0$, then the field distribution on this plane would be given by

$$A_1 \cos\ [kx \sin \theta_1 - \omega t]$$

Similarly, the field (on the plane of the film) due to a plane wave making an angle θ_2 with the z-axis, will be given by

$$A_2 \cos\ [kx \sin \theta_2 - \omega t]$$

The resultant intensity distribution would be proportional to

$$\langle [A_1 \cos\ \{k\,x \sin \theta_1 - \omega t\} + A_2 \cos\ \{k\,x \sin \theta_2 - \omega t\}]^2 \rangle$$

$$= \frac{1}{2}A_1^2 + \frac{1}{2}A_2^2 + A_1 A_2 \cos\ [k\,x\,(\sin \theta_1 - \sin \theta_2)]$$

$$= \frac{1}{2}(A_1 - A_2)^2 + 2A_1 A_2 \cos^2\left[\frac{kx}{2}(\sin \theta_1 - \sin \theta_2)\right]$$

For $A_1 = A_2$, the above expression simplifies to

$$2A^2 \cos^2\left[\frac{kx}{2}(\sin \theta_1 - \sin \theta_2)\right]$$

showing that the intensity remains constant along lines parallel to the y-axis with fringe spacing depending on the values of θ_1 and θ_2. Further, the intensity distribution is of the $\cos^2$ type (cf. Fig. 14.11).

(b) Before we calculate the transmitted field of the hologram, we first consider a narrow slit of width b being illuminated by a plane wave (see Fig. 21.5). Consider an element ds at a distance s from the center of the slit. Then the amplitude at a far away point P due to this element would be proportional to $\sin\ [k\,(r - s \sin \theta) - \omega t]\ ds$; here $k = 2\pi/\lambda$ and θ is defined in Fig. 21.5. Thus the total field in the direction θ would be given by

$$E \approx A \int_{-b/2}^{+b/2} \sin\ [k(r - s \sin \theta) - \omega t]\, ds \qquad \textbf{(21.12)}$$

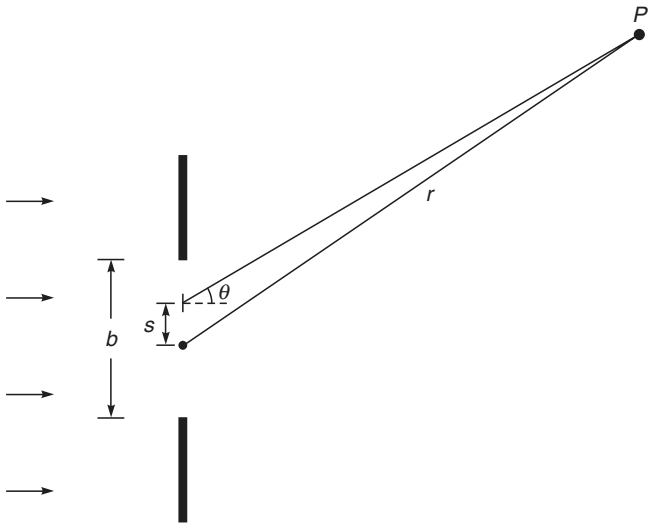


Fig. 21.5 A plane wave incident on a narrow slit of width b.

where A is a constant. The above integral can also be written as

$$E = A \int_{-b/2}^{+b/2} [\sin\,(kr - \omega t) \cos\,(ks \sin \theta) - \cos\,(kr - \omega t) \sin\,(ks \sin \theta)]\, ds$$

$$= 2\,A \sin\,(kr - \omega t)\, \frac{\sin\left(\frac{kb}{2}\sin\theta\right)}{k \sin \theta}$$

where the second integral is zero because the integrand is an odd function of s. Thus,

$$E = Ab \sin\,(kr - \omega t)\, \frac{\sin \beta}{\beta} \qquad \textbf{(21.13)}$$

where

$$\beta = \frac{1}{2}\,kb \sin \theta = \frac{\pi b \sin \theta}{\lambda}$$

which is of the same form as obtained in Sec. 18.2. In the present case, the hologram has a $\cos^2 \alpha s$ type of variation in transmittance and hence the transmitted field will be of the form

$$E = A \int_{-b/2}^{+b/2} \cos^2 \alpha s \, \sin [kr - ks \, (\sin \theta - \sin \theta_i) - \omega t] ds \qquad \textbf{(21.14)}$$

where θ_i represents the angle of incidence of the illuminating plane wave. Thus,

$$\begin{aligned} E = \frac{1}{2} \int_{-b/2}^{+b/2} (1 + \cos 2 \alpha s) \times \\ [\sin (k r - \omega t) \cos \{k s (\sin \theta - \sin \theta_i)\} \\ - \cos (k r - \omega t) \sin \{k s (\sin \theta - \sin \theta_i)\}] \, ds \\ = \frac{1}{2} A \sin (kr - \omega t) \left[\int_{-b/2}^{+b/2} \cos \{ks(\sin \theta - \sin \theta_i)\} \, ds \right. \\ + \frac{1}{2} \int_{-b/2}^{+b/2} \cos \{ks \, (\sin \theta - \sin \theta_i + 2\alpha)\} \, ds \\ \left. + \frac{1}{2} \int_{-b/2}^{+b/2} \cos \{ks(\sin \theta - \sin \theta_i - 2\alpha)\} \, ds \right] \end{aligned} \qquad \textbf{(21.15)}$$

The above integrations can easily be carried out. Thus, for example,

$$\int_{-b/2}^{+b/2} \cos \{ks \, (\sin \theta - \sin \theta_i + 2 \, \alpha)\} \, ds$$

$$= \frac{\sin \left[b \frac{k}{2} (\sin \theta - \sin \theta_i + 2\alpha) \right]}{\frac{k}{2} (\sin \theta - \sin \theta_i + 2\alpha)} \qquad \textbf{(21.16)}$$

which becomes more and more sharply peaked around $\sin \theta = \sin \theta_i - 2\alpha$ as $b \to \infty$, i.e., as the size of the hologram becomes larger. Thus, the three integrals in Eq. (21.15) in the limit of a large value of b give rise to three plane waves propagating along $\sin \theta = \sin \theta_i$, $\sin \theta = \sin \theta_i - 2\alpha$ and $\sin \theta = \sin \theta_i + 2\alpha$, which represent the zero-order and two first order waves.

Example 21.2 Consider the formation of a hologram with a point object and a plane reference wave [(see Fig. 14.13(a))]. Choose the z-axis to be along the normal from the point source to the plane of the photograph, assumed to be coincident with the plane $z = 0$. For simplicity, assume the reference wave to fall normally on the photographic plate. Obtain the interference pattern recorded by the hologram.

Solution: Let the point source be situated at a distance d from the photographic plate. The field at any point $P(x, y, 0)$ on the photographic plate, due to waves emanating from the point object would be given by

$$O(x, y, z = 0, t) = \frac{A}{r} \cos (k r - \omega t) \qquad \textbf{(21.17)}$$

where $r = (x^2 + y^2 + d^2)^{1/2}$ and A represents a constant. A plane wave traveling along a direction parallel to the z-axis would be given by

$$R(x, y, z, t) = B \cos (kz - \omega t) \qquad \textbf{(21.18)}$$

Hence, the field due to the reference wave at the plane of the photographic plate ($z = 0$) would be

$$R(x, y, z = 0, t) = B \cos \omega t \qquad \textbf{(21.19)}$$

Thus, the total field at the plane of the photographic plate would be

$$\begin{aligned} T(x, y, t) &= O(x, y, z = 0, t) + R(x, y, z = 0, t) \\ &= \frac{A}{r} \cos (kr - \omega t) + B \cos \omega t \end{aligned} \qquad \textbf{(21.20)}$$

The recorded intensity pattern would be

$$\begin{aligned} I(x, y) &= \langle | T (x, y, t) |^2 \rangle \\ &= \left\langle \left| \frac{A}{r} \cos (k r - \omega t) + B \cos \omega t \right|^2 \right\rangle \end{aligned} \qquad \textbf{(21.21)}$$

where, as before, angular brackets denote time averaging. Carrying out the above time averaging, we get

$$I(x, y) = \frac{A^2}{2r^2} + \frac{B^2}{2} + \frac{AB}{r} \cos kr \qquad \textbf{(21.22)}$$

If we assume that $d >> x, y$ (which is valid in most practical cases), we can write

$$r = (x^2 + y^2 + d^2)^{1/2} \approx d + \frac{x^2 + y^2}{2d} \qquad \textbf{(21.23)}$$

Thus,

$$I(x, y) = \frac{A^2}{2d^2} + \frac{B^2}{2} + \frac{AB}{r} \cos \left[kd + \frac{k}{2d} (x^2 + y^2) \right] \qquad \textbf{(21.24)}$$

The resultant fringe pattern is circular and centered at the origin (see Example 14.7). The hologram thus formed is essentially a zone plate with the transmittance varying sinusoidally in contrast to the Fresnel zone plate [see Fig. 14.13(b) and Sec. 20.3].

21.3 REQUIREMENTS LO 2

Since holography is essentially an interference phenomenon, certain coherence requirements have to be met with. In Chapter 17, we had introduced the notion of coherence length. Thus, if stable interference fringes are to be formed (so that they are recordable), the maximum path difference between the object wave and the reference wave should not exceed the coherence length. Further, the spatial coherence

is important so that the waves scattered from different regions of the object could interfere with the reference beam.

During reconstruction, the reconstructed image depends both on the wavelength and the position of the reconstructing source. Hence if the resolution in the reconstructed image has to be good, the source must not be broad and must be emitting a narrow band of wavelengths. It may be worthwhile mentioning here that the reconstruction process has associated with it aberrations similar to that in the images formed by lenses. If the reconstruction source is of the same wavelength and is situated at the same relative position with respect to the hologram as the reference source, then the reconstructed image does not suffer from any aberrations.

Another critical requirement in making holograms is stability of the recording arrangement. Thus, the film, the object and any mirrors used in producing the reference beam must be motionless with respect to one another during exposure. One more requirement which is not so obvious (but is a necessity) is the resolution of the film. Two plane waves making angles $+\theta$ and $-\theta$ with the axis, produce an interference pattern with spacing $d = \frac{\lambda}{2 \sin \theta}$. Assuming $\theta = 15°$ and $\lambda = 6328$ Å (He–Ne laser), one obtains $d = 1.222 \times 10^{-3}$ mm; thus the spatial frequency is 818 limes/mm. Thus the photographic plate should be able to record fringes as close as 0.1222×10^{-4} mm apart. This requires special kinds of material which tend to be exceedingly slow, thus taking the stability requirements even further. Some of the holographic materials are 649F Kodak or 10E 75 or 8E 75 Agfa–Gaevert films and plates.

21.4 SOME APPLICATIONS OF HOLOGRAPHY

LO 3

The principle of holography finds applications in many diverse fields.* The ability to record information about the depth finds application in studying transient microscopic events. Thus, if one has to study some transient phenomenon which occurs in a certain volume, then using ordinary microscopic techniques it becomes difficult to first locate the position and make observation. If a hologram is recorded of the scene, then the event gets frozen into the hologram and hence one can focus through the depth of the reconstructed image and study the phenomenon at leisure.

One of the most promising applications of holography lies in the field of interferometry. The ability of the holographic process to release the object wave when reconstructed with a reconstruction wave allows us to perform interference between different waves which exist at different times. Thus, in the technique called **double exposure holographic interferometry**, the photographic plate is first partially exposed to the object wave and the reference wave. Then, the object is stressed and the photographic plate is again exposed along with the same reference wave. The photographic plate after development forms the hologram. When this hologram is illuminated with a reconstruction wave, then two object waves emerge from the hologram; one of them corresponds to the unstressed object and the other to the stressed object. Since the object waves themselves have been reconstructed, they interfere and produce interference fringes. These interference fringes are characteristic of the strain suffered by the body. A quantitative study of the fringe pattern produced in the body gives the distribution of strain in the object.

To understand the formation of the fringe pattern, we assume that the deformation of the object has been such as to alter only the phase distribution. Thus, if

$$O(x, y, t) = A(x, y) \cos [\phi(x, y) - \omega t] \qquad \textbf{(21.25)}$$

represents the object wave (in the hologram plane) when the object is unstressed [see Fig. 21.6(a)] and if $O'(x, y, t)$ represents the object wave when the object is stressed [see Fig. 21.6(b)] then we may write

$$O'(x, y, t) = A(x, y) \cos [\phi'(x, y) - \omega t] \qquad \textbf{(21.26)}$$

where the phase distribution has been assumed to change from $\phi(x, y)$ to $\phi'(x, y)$. On reconstruction, each of the above two object waves emerge from the hologram and what would be observed will be the intensity pattern due to interference of the two waves which would be given by**

$$\begin{aligned} I(x, y) &= \langle [A(x, y) \cos\{\phi(x, y) - \omega t\} \\ &\quad + A(x, y) \cos\{\phi'(x, y) - \omega t\}]^2 \rangle \\ &= A^2(x, y) + A^2(x, y) \cos [\phi'(x, y) - \phi(x, y)] \end{aligned} \qquad \textbf{(21.27)}$$

Thus, whenever

$$\phi'(x, y) - \phi(x, y) = 2m\pi, \; m = 0, 1, 2, \ldots \qquad \textbf{(21.28)}$$

the two waves would interfere constructively and whenever,

$$\phi'(x, y) - f(x, y) = (2m + 1) \frac{\pi}{2}; \; m = 0, 1, 2, \ldots \qquad \textbf{(21.29)}$$

the two waves interfere destructively. Thus, depending on $[\phi'(x, y) - \phi(x, y)]$, one obtains, on reconstruction, the

* See, e.g., Refs. 21.3–21.12.

** The reconstruction process produces other wave components also but as was observed earlier, these components travel along different directions. Here we are concerned only with the object waves.

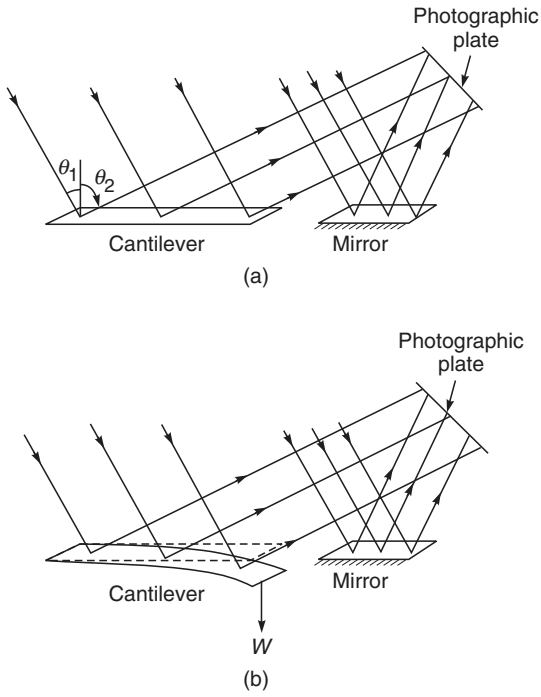


Fig. 21.6 (a) Recording of the unstressed object wave. (b) Recording of the stressed object wave on the same emulsion to produce the doubly exposed hologram.

object superimposed with bright and dark fringes (see Fig. 21.7).

We will consider here a simple application of the above technique in the determination of the Young's modulus of a material. If we have a bar fixed at one end and loaded at the other and if it results in a displacement δ of the end of the bar, then one can show that*

$$\delta = \frac{W L^3}{3Y I} \qquad \textbf{(21.30)}$$

where W is the load, L is the length of the bar, I is the moment of inertia of cross section which for a rectangular bar of width a and thickness b, is given by $I = ab^3/12$; Y represents the Young's modulus of the material of the rod. Thus if we could determine δ for a given load, then Y can be determined from Eq. (21.30).

We will first determine an expression for $(\phi' - \phi)$. In Fig. 21.6, we have shown the undisplaced and displaced positions of the cantilever illuminated by a laser light along a direction making an angle θ_1 with the z-axis. We observe the cantilever along a direction making an angle θ_2 with the z-axis. The phase change when the cantilever undergoes a displacement δ as shown in Fig. 21.6(b) would be

$$\phi' - \phi = \frac{2\pi}{\lambda}(\delta \cos\theta_1 + \delta \cos\theta_2)$$

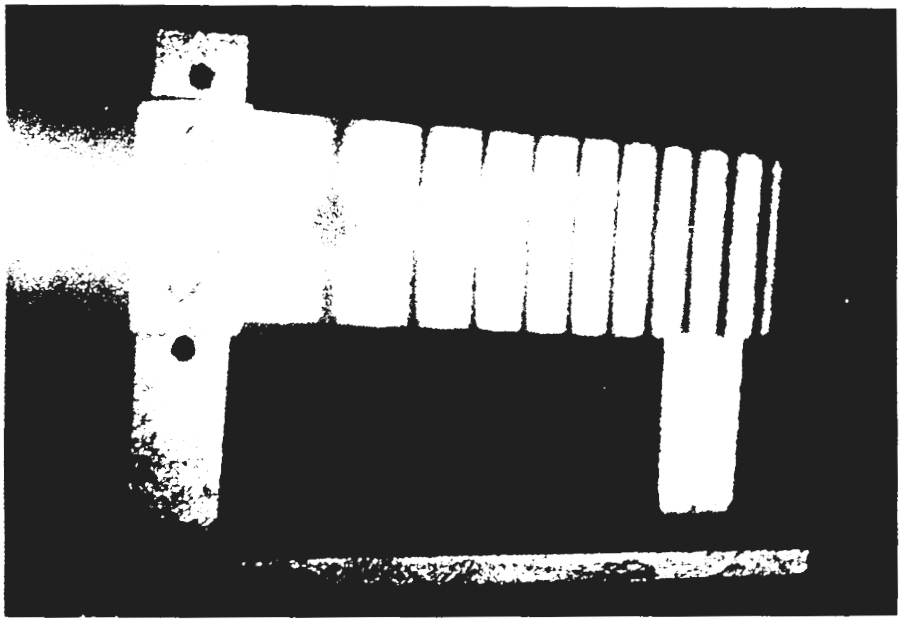


Fig. 21.7 Interference fringes produced in the measurement of Young's modulus using double exposure interferometry. [*Photograph courtesy: Professor R. S. Sirohi*].

* See, e.g., Ref. 21.13, p. 75.